The Future of Feedback: Integrating Peer and Generative AI Reviews to Support Student Work*

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Abstract

This paper explores the integration of generative artificial intelligence (AI) in education to enhance feedback processes and improve learning experiences. The main goal of the study is to investigate the potential of generative AI for feedback, specifically in complementing peer feedback practices among graduate students enrolled at a US-based university during the 2023 academic term. Drawing on existing literature, the study examines the application of generative AI and its implications for feedback mechanisms. Employing an exploratory research design, the study gathers both quantitative and qualitative data through post-course surveys to address key research questions regarding the quality, usefulness, and actionability of peer and AI reviews, as well as their respective advantages and disadvantages. Findings indicate that peer reviews were consistently perceived slightly higher across all three dimensions compared to AI reviews, with thematic analysis revealing the unique strengths and limitations of each review type. This research underscores the importance of integrating human expertise with AI technology in feedback mechanisms, offering practical insights for educators, instructional designers, and policymakers seeking to enhance feedback experiences through emerging digital technologies.

Keywords: Artificial Intelligence (AI), Generative AI, Feedback, Peer Review, Higher Education

Background and Rationale

Assessment, in its various forms, serves as a means of evaluating student performance and understanding. Traditionally associated with summative evaluations, assessments are often reported at the end of a course of study, providing a snapshot or summarization of student achievements (Sadler, 1989) or the overall value of an educational program (Scriven, 1967). While summative assessments offer valuable insights into student proficiency, they often fall short in providing actionable feedback for ongoing improvement (Black & William, 1998; Taras, 2008). Formative assessment, however, unlike its summative counterpart, transcends evaluation towards the end of the learning cycle, and instead focuses on continuous feedback, as an integral part of the instructional process itself (Garrison & Ehringhaus, 2007). It is the antiseptic version of assessment (Scriven, 1967), wherein teacher gathers information or data on what their students know and adapt their pedagogical approaches accordingly (Cope & Kalantzis, 2015; Gotwals et al., 2015). According to Bloom (1969), the essence of formative

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assessment resides in its fundamental objective, which is to offer feedback at every phase of the teaching-learning process.

Despite the fundamental principle of formative assessment emphasizing real-time, high-quality feedback during the learning process (Black & William, 2006; Cope & Kalantzis, 2015; Mahanan et al., 2021) with evidence that suggests its positive impact on student achievement (Black & William, 1998, 2006; Shepard, 2010), its scope and utilization often face limitations (Garrison & Ehringhaus, 2007; Torrance, 2012). Further, research indicates that the actual implementation of formative assessment in educational practice falls short of expectations (Mahanan et al., 2021; Shepard, 2010). What's more, many of the studies supporting the efficacy, had severely flawed research designs yielding uninterpretable results (Kingston & Nash, 2011). Additionally, sometimes it takes forms that could be termed "pseudo-formative" wherein assessments are ostensibly labeled as formative but effectively operate in a summative manner. This could include frequent select response assessments (regular quizzes or multiple-choice tests) that can provide valuable insight into students' understanding of course material, but they may lack the depth and specificity necessary for effective formative feedback, and instead emphasize root memorization or a retrospective judgment of each learner's capacity to recall (Cope et al., 2020) as well as the quantification of performance through numerical scores (Simonson et al., 2000; Vojak et al., 2011). Another example of pseudo- formative assessment tools that is getting progressively more common these days are automated grading systems powered by natural language processing, designed to assess written assignments or responses provided by students using computational linguistics techniques that generate numerical ratings and/or automated feedback (often surface-level) without offering specific, actionable suggestions (Cope et al., 2011; Gamage et al., 2021; Vojak et al., 2011). Torrance's critique of these pseudo-formative assessments aptly characterizes them as "conformative" at best and "deformative" at worst (Torrance, 2012). By shedding light on the disparity between the professed formative nature of these assessments and their actual effects on student learning, Torrance underscores the importance of reevaluating assessment practices to ensure they authentically serve the intended purpose of nurturing students' ongoing development.

Within the framework of an assessment-driven education system, there exists a notable hesitancy to deviate from established methods and practices that appear to yield satisfactory results (Kopparla & Saini, 2022). However, Garrison and Ehringhaus (2007) highlight that the traditional feedback systems while historically instrumental in furnishing students with valuable performance insights, often face challenges in scalability, time efficiency, and personalized guidance. A noteworthy attempt to address these limitations is observed in the implementation of peer-to-peer assessment, wherein learners evaluate and furnish feedback on each other's work (Brún et al., 2022). Peer-to-peer assessment is an excellent example of crowdsourcing, a general practice of obtaining information or input into a task soliciting the efforts of a large number of people (Surowiecki, 2004) reducing the time and workload of teachers (Li et al., 2022) and allowing them to concentrate their efforts on other learning and pedagogical issues (Double et al., 2020; Gamage et al., 2021). This kind of feedback allows course instructors to enhance their courses (Gamage et al., 2021) without the manual labor of grading such exercises given the sheer number of learners, workload, and social interactions (Gamage et al., 2015), particularly when automated grading (or popular system – LMS that uses or claims to employ artificial intelligence) is not yet reliable in these contexts (Sánchez-Prieto et al., 2020, as cited in Gamage et al., 2021). Moreover, engagement monitoring features in LMSs,

often utilized for disciplinary purposes to track students perceived as falling behind –who are not “keeping up”, may deviate from genuine formative intent (Cope & Kalantzis, 2024b).

As we embrace the digital era and leverage the capabilities of generative AI models, there emerges a unique opportunity to further revolutionize the feedback process, offering more comprehensive, timely, and tailored evaluations. While peer review can facilitate valuable interpersonal interactions that significantly contribute to motivation, emotional support, and the cultivation of a growth-oriented mindset, the integration of generative AI reviews offers the advantage of rapid analysis, automated grading, and the ability to process vast amounts of data, ensuring timely feedback for every student driven by thorough prompts in the backend. This integration presents a forward-looking approach to feedback processes, fostering an environment where technological advancements harmonize with human intuition to enhance student learning experiences. However, as educational institutions increasingly adopt AI-driven feedback mechanisms, understanding how participants perceive the utility of AI reviews in comparison to peer reviews becomes paramount. Furthermore, understanding how participants distinguish the utility of AI-generated feedback from peer reviews can provide an overview of the evolving dynamics of technology-mediated learning experiences. Likewise, insights into how participants' emotional responses towards AI influence their reception of feedback can contribute to the nuanced understanding of the human-AI relationship in an educational context especially since individuals' feelings towards AI systems have been identified as pivotal factors shaping their overall experiences (Reeves & Nass, 1996).

By combining the expertise and empathy of peer review with the computational power and data-driven insights of AI models, the present study makes a case for a dynamic and symbiotic feedback ecosystem facilitated by the emerging sub-disciplinary areas within the learning sciences: educational data mining and learning analytics. This paper is part of a broader research endeavor where Principal Investigators (PIs) and Co-PIs utilize both structured (information that is organizable in a predefined format) and unstructured data (lacks a predefined data model or organizational schema) to furnish student feedback, track progress, and conduct summative assessments using an in-house analytics application (Cope & Kalantzis, 2019, 2023a). This study aims to explore and harness the potential of merging peer and generative AI feedback in the educational landscape. The purpose of this study is to contribute to these ongoing discussions by examining university students’ opinions and self-reported experiences with peer and AI feedback. Specifically, in this work, we explore graduate students’ views of peer reviews in comparison to those offered by an AI chatbot. The insights gleaned promise to inform pedagogical practices, offering a holistic approach that amalgamates the strengths of both human and machine-generated feedback.

Literature Review

Assessment has always been an integral part of education. Concurrently, the pursuit of effective review mechanisms to foster student learning and growth has been deeply rooted in our educational practices. This literature review delves into the exploration of the potential of generative artificial intelligence (AI) in conjunction with peer review practices, focusing on how this amalgamation could revolutionize the feedback landscape.

Peer Feedback

Feedback is essentially information provided by any actor (e.g., teacher, peer, book, parent, self, experience) with respect to one's performance or understanding (Hattie & Timperley, 2007). These actors are often regarded as a central component of educational practice (Double et al., 2020). Peer feedback (also called peer review[3]) is the quantitative evaluation and qualitative feedback of a learner's performance by another learner (Patchan & Schunn, 2016, as cited in Patchan et al., 2022). In simpler words, students provide feedback reviews on their peers' work and receive feedback reviews on their own work (Nicol et al., 2014). It is important to note that the terms peer review, peer feedback, peer grading, and peer assessment have been used interchangeably throughout the literature (Wilson et al., 2015, as cited in Vander-Schee et al., 2022). Peer feedback is said to be a social process (Panadero & Alqassab, 2019) that has a pivotal position in social development theory to facilitate learning and social interaction (Vander-Schee et al., 2022).

The Senior Editorial Assistant of F1000, Yousuf Al-Mousawi in his article (Al-Mousawi, 2020) provided a brief about the evolution and history of peer feedback – both the trace of the concept as well as the beginning of the actual process of peer feedback in practice. According to him, the concept (and not the actual process) of peer review, dates to ancient Greece around the 5th Century BC following which he mentions the book by a Syrian author called 'Ethics of the Physician' that holds the first documented description of a peer-feedback process in it. Furthermore, he highlighted various accounts and activities starting from Henry Oldenburg, the first editor of the 'Philosophical Transactions' (1665), which is claimed by some as the first record of an editorial pre-publication peer-feedback to the Royal Society of Edinburgh's 'Medical Essays and Observations' in 1731 that is considered to be the first peer-reviewed publication to distilling the nuances of rubrics, referees (external and internal) along the way to finally the middle of 20th century with journals such as 'Science' and 'The Journal of the American Medical Association (JAMA)' that shaped the 'open peer review' that we and many academic journals subscribe to now.

The term 'Peer Assessment' is said to have been proposed by Keith Topping in 1988 (Chien et al., 2019) in the article titled 'Peer Assessment Between Students in Colleges and Universities', where peer assessment was defined as a strategy that engages students to enhance their higher-order thinking. Numerous studies have reported on the benefits of employing peer assessment, highlighting, for example, significant academic achievements (Chien et al., 2019; Gamage et al., 2021; Double et al., 2020; Hattie & Timperley, 2007; Huertas-Abril et al., 2021; Panadero & Alqassab, 2019) and behavioral change (e.g., boost in confidence, engagement, etc.) (Brún et al., 2022; Vander-Schee et al., 2022). This type of feedback has also been deemed a means to enhance students' learning through knowledge diffusion and the exchange of ideas (Gamage et al., 2015; Li et al., 2022; Nicol et al., 2014). In addition, the utility of peer assessment extends beyond mere performance augmentation. It is instrumental in equipping students with multifaceted proficiencies, encompassing the art of critiquing, structuring, evaluating, rating, and furnishing feedback (Nicol et al., 2014) as well as helping students learn from mistakes (Huertas-Abril et al., 2021) and impart other important skills (e.g., critical thinking, analytical thinking, etc.) in the process (Double et al., 2020). The pedagogical value of

[3] In this paper, human reviews, peer reviews, or human/peer reviews has been used synonymously. The same applies to reviews and feedback.

peer assessment becomes increasingly apparent as it nurtures skills integral to self-assessment and enhances presentation capabilities (Nikolic et al., 2018).

Hattie and Timperley (2007) delve into more effective feedback that should stimulate cognitive thinking and should be directed towards actionable steps for improvement facilitating the process of bridging the gap between where the students are now and where they need to be. They put forward what is currently the most cited framework, backed by pedagogical and psychological literature on feedback (Panadero & Lipnevich, 2022). The authors outline three strategic approaches centered around core questions for effective student feedback: feed-up (Where am I going?), feed-back (How am I going?), and feed-forward (Where to next?). Feed-up sets clear goals and expectations; Feed-back evaluates progress, recognizing strengths and areas for improvement; and Feed-forward provides specific strategies for enhancing learning outcomes. They suggested that these questions offer a theoretical framework (Figure 1) for understanding how different types of feedback can be used across four levels: task, process, self-regulation, and self-level (Otaki, 2023; Panadero & Lipnevich, 2022).

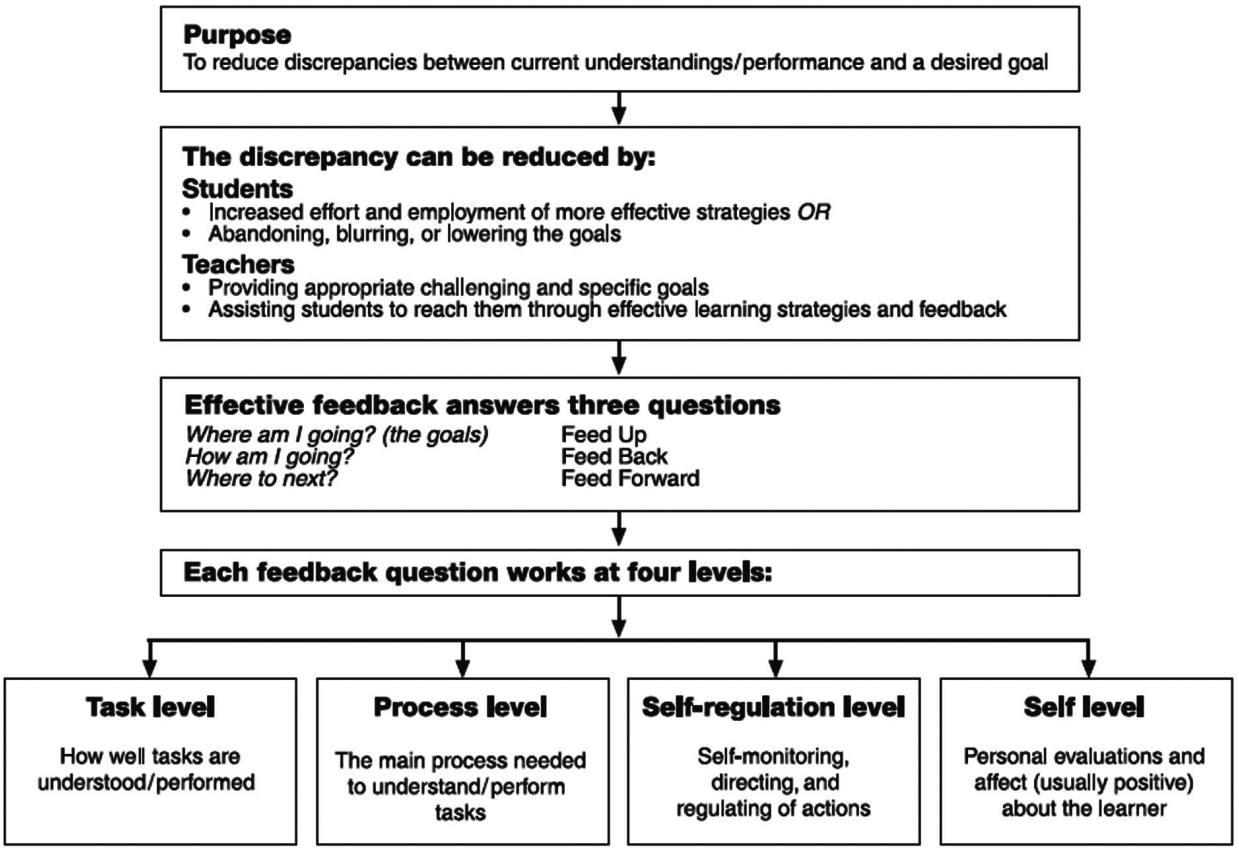

Figure 1: The power of feedback (Hattie & Timperley, 2007, p. 87).

Further, studies have reported peer assessment as a tool to enhance learner interaction (Hattie & Timperley, 2007; Gamage et al., 2021), increase learner motivation and encourage them to work harder (Chien et al., 2019), and can be regarded as a reliable and valid measure (El-Senousy, 2020; Patchan et al., 2022). Especially, if anonymity is involved, the gains go multifold (Panadero & Alqassab, 2019). Scholarly investigations underscore the assertion that peer evaluations and feedback, when administered anonymously, yield significant enhancements in learning performance and learners' perceptions of the learning process (Hattie & Timperley, 2007; Li et al., 2019; Panadero & Alqassab, 2019). At the same time,

anonymous feedback minimizes favoritism or bias while being accompanied by constructive feedback (Panadero & Alqassab, 2019). A comprehensive meta-analysis by Li et al. (2019) encompassing 58 distinct research studies, each conforming to an experimental or quasi-experimental design and encompassing both control and experimental groups, within the temporal span from 1950 to 2017, yielded a substantive observation. This analysis determined that students who abstained from participation in peer assessment and those who engaged in peer assessment exhibited a noteworthy .291 standard deviation increment (with a 95% confidence interval ranging from .194 to .388) in their academic performance. Furthermore, in the meta-analysis, there were more positive effect sizes than negative effect sizes meaning peer assessment, in general, has a nontrivial positive effect on students' learning performance. The researcher also found that the computer- mediated peer assessment provides greater learning gains compared to the paper-based peer assessment.

Likewise, some claim that peer feedback offers positivity and mutuality of learning in accordance with the principles of collaborative learning and social constructivism (El-Senousy, 2020), which is beneficial for both assessment receivers and peer assessors (Li et al., 2022; Vander-Schee et al., 2022) as it has the potential to improve individual skills, teamwork skills, and practical performance (El-Senousy, 2020). A quasi-experimental study published in 2019 by Chien et al (2019) examined the effects of peer assessments on English-speaking performance, anxiety, motivation, and critical thinking in a spherical video-based virtual reality (SVVR) environment using Analysis of Covariance (ANCOVA). The participants of this study were 69 Taiwanese high school students (average age 16) with 36 students in the control group and 33 in the experiment group. The study relied on individual interview results and overarchingly reported positive effects for peer-assessment-based SVVR approach compared with the non-peer-assessment-based SVVR approach in terms of the learners' English speaking [with $F(1,66) = 5.63$ for $p = 0.04 < 0.02$], learning motivation [with $F(1,66) = 5.50$ for $p = 0.02 < 0.05$], and critical thinking skills [with $F(1,66) = 7.97$ for $p = 0.006 < 0.01$], as well as reducing learning anxiety [with $F(1,66) = 5.01$ for $p = 0.03 < 0.05$]. Additionally, the students who had participated in the peer-assessment-based SVVR approach performed significantly better than those who had used the non-peer- assessment-based SVVR approach in all six dimensions of the adapted Rubric of English- speaking performance.

Huertas-Abril et al. (2021) conducted a mixed-method analysis of the responses of 58 pre-service primary bilingual teachers at the University of Córdoba (Spain). The results indicated a significant interest in and positive attitude towards peer assessment as a means to enhance performance. Similarly, Brún et al. (2022), in a quantitative descriptive study aimed at evaluating the expectations and perceived utility, acceptability, and satisfaction with peer assessment reported high satisfaction scores for peer assessment as an alternate assessment method. What's more, according to a study conducted on two sections of online courses (n = 159 and n = 75 respectively) at the same university with different formats (asynchronous and synchronous) hosted on Zoom (video conferencing tool) found no significant difference in instructor and peer-assigned grades which is consistent to the research findings of Li et al. (2022) and Patchan et al. (2022).

However, just like any other tool or strategy, it is imperative to be mindful of risks, biases, barriers, and implementation issues that are associated with peer-to-peer assessment in different teaching-learning environments (Li et al., 2019). There is a need to explain the expectation of engaging in a peer review process (Nicol et al., 2014) and familiarize students

with the process to avoid apprehension, perceiving the task as daunting, feeling less confident (Nicol et al., 2014; Panadero & Alqassab, 2019), and doubting their ability to provide valuable and accurate feedback (Brún et al., 2022). Hattie and Timperley's (2007) study posits that the type of feedback provided and the way it is given can play a crucial role. Moreover, Brún et al. (2022) found a mixed result regarding whether peers can assess fairly (M = 3.38, SD = 1.16) – the lowest scoring item, and the item with the highest standard deviation. This has been supported by evidence showcased in the works of Patchan et al. (2022), where peer reviewers provided more inaccurate feedback, while the teacher's feedback was always accurate. A quasi-experimental study conducted by Nikolic et al. (2018) over five teaching semesters, spanning two and a half years beginning in July 2013. The study investigated the differences between traditional delivery, self-assessment, and a combined approach of self-assessment with peer feedback. The results revealed no significant difference between the control and experimental groups, indicating that there was no notable improvement compared to traditional methods, despite students' perceptions that peer assessment was beneficial for their learning. Besides, discrepancies have been noted in peer feedback processes, particularly when involving participants with mixed levels of proficiency. A study conducted by Lu and Law in 2012 brought to light that weaker students may not derive the same level of benefit from peer feedback as their stronger counterparts. This disparity is often attributed to the limited explanatory content within peer review comments and the challenges faced by students in interpreting such feedback. Consequently, weaker students may exhibit reduced engagement with and implementation of peer feedback recommendations. It is worth noting that the advantages of peer feedback may accrue to the provider rather than the recipient. For students at lower proficiency levels, a focus on surface-level aspects may limit the extent to which they can truly benefit from the peer feedback process.

Generative AI (GenAI or GAI)

The education sector has historically been a little slow to embrace new technologies (Nobaew, 2021), however, the COVID Pandemic forced us to broaden our teaching-learning practices using digital technologies and has future-forwarded pedagogical practices by decades (Baidoo-Anu & Owusu Ansah, 2023; Nobaew, 2021). Notably, in the past two years, educators have embraced technologies such as AI, virtual reality, etc., either individually or in conjunction with other technologies and systems, to enhance and reimagine teaching processes. This momentum, spurred by the pandemic, is anticipated to make the utilization of these resources a preferred choice for online higher education in the foreseeable future (Rangel-de Lázaro & Duart, 2023). Particularly, AI, or 'artificial intelligence' as described by Wang et al. (2018), is a simulation of human thinking and consciousness that can be perceived as a social agent (Reeves & Nass, 1996). AI grants machines the capacity to think and create independently, akin to endowing them with the cognitive abilities of their human creators, a concept that has moved beyond science fiction and into our daily lives (Sharma, 2023) and is growing by the day (Cope et al., 2020). Nguyen et al. (2022) characterize AI as "one of the most pivotal developments of the century". The term AI was officially coined in 1956 by Marvin Minsky and John McCarthy who are considered as its founding fathers (Haenlein & Kaplan, 2019).

However, the foundational work in this field is often credited to Alan Turing who worked and developed a code-breaking machine called 'The Bombe' for the British government during World War II, and these experiences prompted him to contemplate machine intelligence. In 1950, he published his seminal article "Computing Machinery and Intelligence" outlining the

creation and assessment of intelligent machines. The enduring legacy of his work is the Turing Test, a benchmark for gauging the intelligence of artificial systems (Haenlein & Kaplan, 2019). If a machine engages in human-like interactions to the point of being indistinguishable from a human, it is deemed intelligent (Haenlein & Kaplan, 2019; Turing, 1950), a proposal in line with ideas echoed in the seminal work of Reeves & Nass (1996). Further, expanding on machine learning (ML), a specialized subset of AI operates on the principle of reverse training, wherein data evaluation and extraction ensue from exposure to labeled examples (von Ende et al., 2023). This shift found its utility in the development of self-learning computer models, proficiently engaged in a multitude of tasks, ranging from prediction to classification, clustering, and segmentation (Madathil et al., 2023). Recent strides in ML have ushered in a new era of innovative technology. Particularly, generative artificial intelligence (GenAI) (Hu, 2022) stands as a testament to the progress and expansion within the field. Over the past decade, it has been regarded as one of the most successful ML frameworks in the evolution of deep learning (DL), a system's ability to understand complex patterns and structures (Hu, 2022; Li, 2017 as cited in Rong et al., 2022). With the capability to focus more on information integration, reconfiguration of knowledge systems through its own thinking and problem-solving, DL is reshaping the field of AI at an exponential rate (Rong et al., 2022).

The product of these advancements in technology and user experience is GenAI (Bandi et al., 2023). GenAI is essentially an unsupervised or partially supervised machine learning framework, which generates artificial artifacts, leveraging a myriad of statistical and probabilistic techniques (Baidoo-Anu & Owusu Ansah, 2023; Hu, 2022). Its utilization of these techniques not only underscores its innovative capabilities but also highlights its potential to contribute significantly to the creation of diverse and meaningful artifacts using existing digital content including but not limited to text, graphics, audio, and video by processing corpus data (Cope & Kalantzis, 2023b; Hu, 2022), by learning existing patterns and distribution (Hu, 2022) as well as by simulating the thinking process of the human brain and finally responding to instructions (Rong et al., 2022). Evidently, such systems surpass human capabilities, as unlike the human brain, they exhibit superior computational power and memory capacity, enabling it to process information at an unparalleled efficiency (Gong, 2021).

Gen AI, specifically ChatGPT, has become a 'cultural sensation' (Thorp, 2023), a remarkable technological advancement that sprang attention to GenAI (Cope & Kalantzis, 2023c) that performs a variety of tasks from any specified prompt and produces human-like responses and discussions on any terminology or topic (Aydin & Karaarslan, 2023; Thunström, 2022). Launched by OpenAI in November 2022, ChatGPT garnered a staggering one hundred million users within a mere two months of its launch, surpassing the adoption rate of any previous product or platform (Cope & Kalantzis, 2023c). The GPT in ChatGPT stands for Generative Pretrained Transformer, a type of AI model that uses natural language processing (NLP) trained to read and generate text often indistinguishable from human-like text responses (Aydin & Karaarslan, 2023; Nisar & Aslam, 2023; Thunström, 2022). Consequently, ChatGPT is an advanced NLP model designed to generate and express new content or ideas in real-time (Aydin & Karaarslan, 2023; Qadir, 2023) using a coherent conversational style and maintaining a consistent persona or identity throughout an interaction (Qadir, 2023). This is essentially possible because of two key technologies: a "Chatbot" and a vast corpus of digital content (as mentioned earlier), often referred to as a "Large Language Model (LLMs)," which the chatbot can interact with (Cope & Kalantzis, 2023c).

The concept of computers as potentially human-like conversational agents dates back to the fundamental question posed by Alan Turing in 1951: "Can computers think?" (Aydin & Karaarslan, 2023). Chatbots are essentially virtual assistants capable of providing appropriate information, answering questions, or performing specific tasks (Clarizia et al., 2018). The foundational concept of chatbots revolves around a structured exchange between a human and a machine, typically operating within the framework of a prompt-and-response dialogue (Tzirides et al., 2023a). The earlier work in this domain can be traced back to the development of the first computer chatbot, known as ELIZA, programmed to engage in conversations that resembled those held with psychotherapist Carl Rogers, which emerged between 1964 and 1966 under the guidance of Joseph Weizenbaum at the Massachusetts Institute of Technology (MIT) (Cope & Kalantzis, 2024a; Tzirides et al., 2023a). On the other hand, LLMs are simply models that are trained on massive datasets, enabling them to understand, generate, and respond to text-based queries (Peng et al., 2023) that are 'grammatically correct and semantically meaningful' (Brynjolfsson et al., 2023, p. 4).

With its launch, ChatGPT has led to a surge in AI tools aimed at creating and developing a variety of applications (digital images, video, audio, text, or code) for different sectors (Aydin & Karaarslan, 2023; Hu, 2022) including chatbot responses, generating ideas for stories and content, and even creating music (Qadir, 2023). In the realm of education, the integration of tools like ChatGPT bears significance for both students and educators. These AI-powered tool finds utility in diverse educational facets, extending beyond the confines of traditional roles (Baidoo-Anu & Owusu Ansah, 2023; Qadir, 2023; Sharma, 2023). With its potential application in 'personalized learning', 'automated essay grading', 'language translation', 'interactive learning', 'adaptive learning' and more, ChatGPT has the capabilities to solve problems and achieve goals (Baidoo-Anu & Owusu Ansah, 2023; Qadir, 2023). Educators, for instance, can leverage its capabilities to craft a wide array of educational materials (such as quizzes, grading, rubric design, etc.) to support and improve pedagogical and assessment practices, (Baidoo-Anu & Owusu Ansah, 2023; Qadir, 2023; Thunström, 2022). Conversely, students can harness its potential to support their learning by seeking assistance with systematic feedback, language translation, essay composition, and more (Baidoo-Anu & Owusu Ansah, 2023; Nisar & Aslam, 2023; Qadir, 2023; Thorp, 2023). Zhai's (2023) study suggests that these systems can contribute to addressing challenges within science education, particularly related to creating performance-based assessments, assessing student progress, delivering constructive and personalized feedback, and providing guidance and recommendations of supplementary materials. A recent study by Demszky and colleagues (2023) that examined the impact of an automated feedback tool on both instructors and students highlighted the significance of the feedback tool for instructors seeking to enhance their teaching practices. The results of this randomized controlled trial, conducted within an online computer science course involving a sample of 1,136 instructors revealed that the tool significantly enhanced instructors' engagement with students' contributions by 13%, leading to improved assignment completion rates and increased overall satisfaction with the course.

Since ChatGPT's introduction to the public in 2022, the academic community has engaged in extensive discourse regarding its potential implications for various stakeholders, including educators, students, and policymakers (Aydin & Karaarslan, 2023; Baidoo-Anu & Owusu Ansah, 2023; Zhai, 2023). Particularly, the AI-generated feedback or GenAI reviews has ignited significant discussions and debates within these circles (Baidoo-Anu & Owusu Ansah, 2023; Dai et al., 2023; Otaki, 2023) as it can use NLP to analyze work and provide feedback based

on predefined criteria (Clarizia et al., 2018; Wongvorachan & Bulut, 2022). This development aligns with Hattie and Timperley's (2007) model, offering a means to solicit continuous personalized feedback more easily for students (Otaki, 2023; Wongvorachan & Bulut, 2022). Moreover, Clarizia et al. (2018) noted that the application of these NLP techniques excels in delivering feedback based on predefined criteria, including aspects such as grammar, syntax, and content which students find straightforward and efficient. Studies have showcased that generative models trained on a dataset of human-graded essays could efficiently grade essays written by students (; Kim, Park, & Lee, 2019 as cited in Baidoo-Anu & Owusu Ansah, 2023; Otaki, 2023; Tzirides et al., 2023a) with a certain degree of accuracy (Mizumoto & Eguchi, 2023). Research suggests that the GenAI reviews are more critical (Tubino & Adachi, 2022; Tzirides et al., 2023a), readable (Dai et al., 2023; Tzirides et al., 2023a), personalized (Brynjolfsson et al., 2023; Chan & Hu, 2023; Cotton et al., 2023; Farrokhnia et al., 2023; Norreton & Schörling, 2023; Zhai, 2023), objective (Tzirides et al., 2023a), consistent (Baidoo-Anu & Owusu Ansah, 2023; Dai et al., 2023; Mizumoto & Eguchi, 2023; Tubino & Adachi, 2022; Tzirides et al., 2023a; Zapata et al., 2023), logistical effective (Dai et al., 2023; Mizumoto & Eguchi, 2023; Norreton & Schörling, 2023; Tzirides et al., 2023a; Zapata et al., 2023), convenient (Mizumoto & Eguchi, 2023; Tzirides et al., 2023a), quicker (Baidoo-Anu & Owusu Ansah, 2023; Cotton et al., 2023; Mizumoto & Eguchi, 2023; Norreton & Schörling, 2023; Tubino & Adachi, 2022; Zapata et al., 2023), and ubiquitous (Cotton et al., 2023; Dai et al., 2023; Tzirides et al., 2023a) as compared to peer feedback. In a study conducted by Seo et al. (2021), it was evident that instructors unanimously recognized the value of immediate feedback facilitated by AI in aiding student learning. This real-time feedback mechanism has proved especially advantageous during periods when instructors might be unavailable, thereby enabling asynchronous learning. AI reviews can also facilitate rapid communication and promote more meaningful interactions between instructors and students. Importantly, these studies have concluded that AI has the capacity to normalize peer review grades by considering each student's historical data on peer reviews with those of their peers from previous iterations of the course.

Cotton et al. (2023) and Zhai (2023) highlight AI's capability to promptly grade assignments while simultaneously delivering personalized feedback, thereby enhancing the efficiency and tailoring of learning experiences. Specifically, in text-based courses, GenAI emerges as a valuable tool for the automatic grading of assignments (Cotton et al., 2023). Mayer et al. (2023) emphasize that GenAI ratings can achieve levels of accuracy comparable to those produced by human assessors aligning with the insights provided by Reeves & Nass (1996). Their study suggests that GenAI can complement peer ratings by providing supplementary insights and evaluations. Similarly, Norreton and Schörling's (2023) study demonstrated that AI-generated feedback often surpasses human-generated feedback, showcasing the promise of AI as a means to surmount the hurdle of crafting feedback. Furthermore, Dai et al.'s (2023) recent study that compared GenAI reviews and instructor feedback on 103 student data science projects in a postgraduate university program revealed that GenAI reviews help educators to provide high-quality personalized feedback consistently even to a large class size. The authors of the study also noted that the feedback provided was characterized by increased readability, consistency, and specificity. These improvements resulted in higher agreement scores, demonstrating significant enhancements in both precision and recall metrics across all five rubric elements. In their recent study, Tzirides et al. (2023a) reported that the GenAI consistently provided more extensive feedback, with an average length of 1335.5 words per review, compared to human peers who averaged 335.5 words. Their findings revealed that

GenAI has the ability to offer more comprehensive and in-depth feedback to students, encompassing a broader range of evaluative perspectives suggesting a more objective approach to evaluating student performance. What's more, they observed that the consistency exhibited by GenAI reviews extends to the sentiment measure similar to the findings of Brynjolfsson et al. (2023). Notably, their findings indicated that GenAI reviews surpassed human-authored reviews in terms of academic language readability. Mizumoto and Eguchi's (2023) study which analyzed 12,100 English essays authored by individuals from 11 distinct linguistic backgrounds between 2006 and 2007, revealed that GenAI significantly reduced the time required for grading, ensured consistency in scoring, and was able to provide immediate scores and feedback, echoed on a similar vein. Based on these findings, they proposed that AI as an automated essay-scoring tool holds the promise of substantially reducing the workload of teachers engaged in the burdensome task of correcting student writing while mitigating evaluation bias and it enables practitioners to concentrate on other crucial aspects of instructions similar (see also Cotton et al.,2023; Norreton & Schörling, 2023; Otaki, 2023; Seo et al., 2021; Tubino & Adachi, 2022).

Following the promise come the challenges and hiccups, and it is important to remember that even though GenAI, as well as ChatGPT and various other AI language models, has revolutionized natural language processing tasks with its ability to generate coherent and contextually relevant text, it is not perfect and there may be instances where it makes errors or offers inaccurate information (Qadir, 2023). The utilization of GenAI reviews for providing feedback and scoring remains a relatively new and actively investigated area of research (Baidoo-Anu & Owusu Ansah, 2023; Otaki, 2023; Tzirides et al., 2023a). There exists a significant gap in our understanding of the patterns, challenges, and potential applications of such AI models (Nisar & Aslam, 2023), especially considering the early stages of GenAI, and these aspects deserve further scrutiny (Brynjolfsson et al., 2023). While AI holds promise in providing immediate, general feedback, there are evident areas for growth and research (Otaki, 2023). For example, these areas encompass algorithmic bias (Baidoo-Anu & Owusu Ansah, 2023; Brynjolfsson et al., 2023; Otaki, 2023; Zhai, 2023), lack of empirical verification (Cope & Kalantzis, 2024a), dependency on training data (Aydin & Karaarslan, 2023; Baidoo-Anu & Owusu Ansah, 2023; Brynjolfsson et al., 2023; Cope & Kalantzis, 2024a; Lindsay et al., 2023; Norreton & Schörling, 2023), prompt formulation (Cotton et al., 2023; Farrokhnia et al., 2023; Mizumoto & Eguchi, 2023; Qadir, 2023), lack of contextual understanding (Baidoo-Anu & Owusu Ansah, 2023; Cope & Kalantzis, 2024a; Farrokhnia et al., 2023; Otaki, 2023), and lack of explainability (Chan & Hu, 2023; Lindsay et al., 2023; Mizumoto & Eguchi, 2023; Zhai, 2023). Additionally, GenAI feedback may not be appropriate for all assignment types (Dai et al., 2023), or be considered less nuanced (Otaki, 2023), vague and generalized (Tzirides et al., 2023a; Zapata et al., 2023), lacking human touch and emotional intelligence (Baidoo-Anu & Owusu Ansah, 2023; Otaki, 2023; Zapata et al., 2023), reliability and consistency (Cope & Kalantzis, 2024a; Norreton & Schörling, 2023; Qadir, 2023). Since GenAI has also been shown to be prone to hallucination or missing references (Baidoo-Anu & Owusu Ansah, 2023; Brynjolfsson et al., 2023; Nisar & Aslam, 2023; Peng et al., 2023), it might offer incorrect answers or misinformation (Aydin & Karaarslan, 2023; Baidoo-Anu & Owusu Ansah, 2023; Norreton & Schörling, 2023; Thorp, 2023). Other concerns involve data privacy (Baidoo-Anu & Owusu Ansah, 2023; Chan & Hu, 2023; Cope & Kalantzis, 2024a; Otaki, 2023), ethics, plagiarism, and academic integrity (Aydin & Karaarslan, 2023; Chan & Hu, 2023; Cope & Kalantzis, 2024a; Cotton et al., 2023; Farrokhnia et al., 2023; Mizumoto & Eguchi, 2023; Otaki, 2023; Thorp, 2023). Clearly, there is a need for further research. As GPTs and similar

advanced language models continue to evolve, it is imperative to address the emerging limitations and challenges identified through recent research.

Furthermore, Tzirides et al.'s (2023a) research revealed that peer reviews perform better than AI reviews in terms of accuracy and consistency in the rating category as well as in sentiment scoring. Dai et al.'s (2023) study emphasized that GenAI, particularly ChatGPT, could not offer a reliable assessment of student performance compared to the human instructor. The authors suggest that this limitation may arise from the AI not being trained on a dataset comprising student assignments of varying quality and the corresponding standard feedback that would enable a more accurate evaluation of their performance. Consequently, they suggest that further investigation is warranted to gauge the depth at which these AI models can offer relevant feedback and automated grades which was echoed in the findings of Zapata et al (2023) and Farrokhnia et al. (2023) as well as aligns with the suggestions made in Seo et al. (2021). Otaki (2023) recommends both educators and students to be critical consumers of AI feedback and understand its limitations compared to the depth and complexity of human feedback as well as warranted further research into examining student and educator behaviors toward AI feedback and its impact on the effectiveness of the said feedback[4]. Lastly, it is necessary to acknowledge the significant environmental footprint associated with the computational resources consumed during the training and inference processes of GPTs (Patterson et al., 2021; Smith et al., 2023). As the size and complexity of these models continue to expand, the energy expenditure required for training them is expected to escalate (Smith et al., 2023). Therefore, mitigating the environmental consequences of large language models warrants careful consideration in the development and deployment of such technologies.

Methodology

Overview and Background

This work is part of a comprehensive research project in the field of cyber-social research (Tzirides et al. 2023b), aimed at investigating the impact of AI generative technologies in higher education through a social knowledge and learning platform, Common Ground Scholar (CGScholar[5]) (Cope & Kalantzis, 2023a). CGScholar provides a collection of web-based social

[4] On a separate note, research suggest that it is also important to ensure an equitable access to these technologies followed by adequate training and support to employ such tools and interpret produced feedback (Qadir, 2023; Cotton et al., 2023), guided by informed ethical and regulatory frameworks (Sharma, 2023) so that we can have more information and datapoints to take better policy and practice decisions (Chan & Hu, 2023; Otaki, 2023; Sharma, 2023).

[5] According to Tzirides et al. (2023a), CGScholar had nearly 350,000 user accounts and was compatible with LTI-compliant learning management systems in 2023. Operated by Common Ground Research Networks, a not-for-profit public benefit corporation, CGScholar offers certain parts of its software suite for free sign-up and usage, while other components incur a modest licensing fee aligned with self-sustainability principles. Saini (2023) notes that the platform accommodates more than 20,000 articles across English and Spanish, Common Ground Research Networks emphasizes cross-sectorial and interdisciplinary studies to advance social knowledge technologies, with a repository comprising over 200 published books and 90 journals. Cope et al. (2020) explain that the CGScholar platform operates on a freemium model, providing open access to certain features while charging for others to ensure platform sustainability. This arrangement is managed by Common Ground Research Networks, headquartered in the Research Park at the University of Illinois. Use cases for CGScholar include

knowledge applications initially developed as part of the Social Literacy project under the Australian Government's Multicultural Education Program, however, over time, it has evolved into a platform serving as a hub for individuals to convene, exchange ideas, and engage in dialogue (Saini, 2023).

The cyber-social research approach integrates agile research methods inspired by contemporary software development processes with design research methods in education (Tzirides et al. 2023b). This approach involves close collaboration between research and software development teams, with active involvement from students alongside research team leaders. The results of the iterative research-design process were used in the development and refinement of the CGMap[6] tool. CGMap is a tool integrated within the CGScholar platform that serves multiple purposes. It enables the creation of concept maps, connecting ideas and concepts through a network of nodes (Cope et al., 2022; Cope & Kalantzis, 2023a) extensively used in providing blind peer-review, AI review, instructor, and self-review against the same rubric.

The participants for this study were students enrolled in courses offered under the Learning Design and Leadership program at the University of Illinois. These courses, spanning 6 weeks in summer or 8 weeks in fall/spring, were delivered via the CGScholar platform. This study particularly focuses on the students' perceptions of the feedback received on their main projects for the respective courses. The participants enrolled in the courses were tasked with creating a piece of work (main project) on their chosen topic, within the realm of "cutting-edge innovation" in educational theory and practice, which they developed throughout the term. This work underwent a comprehensive evaluation process involving blind peer reviews and AI feedback utilizing the CGMap tool. In addition to providing avenues to create concept maps of ideas and concepts linked together through a network of linked nodes (Cope et al., 2022; Cope & Kalantzis, 2023a) extensively used in the said blind peer-review, CGMap is connected to

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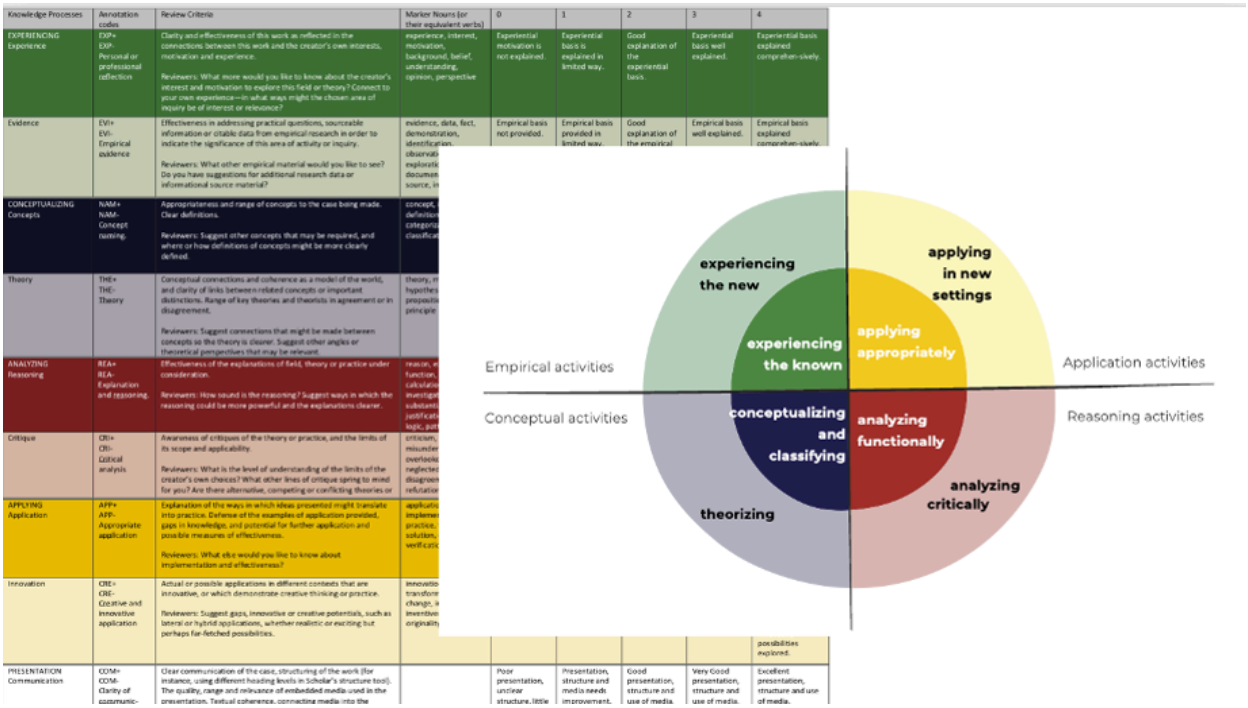
literacy programs spanning grades 4 to 12 in schools, higher education across various disciplines such as education, engineering, medicine, and veterinary medicine, as well as global social learning initiatives led by organizations like the Red Cross and the World Health Organization (Cope & Kalantzis, 2023a; Tzirides et al., 2023a).

[6] Both Cope & Kalantzis (2023a) and Tzirides et al. (2023b) highlights the capabilities of the CGMap tool, which allows users to generate graphical representations of interconnected concepts or ideas, specifically tailored for multimodal scholarly works. Distinguished from conventional concept mapping software, CGMap features a unique split-screen architecture (refer to Fig. 4 or Fig. 5), wherein the left side presents an extended multimodal work, while the right side hosts a concept mapping palette. Further elucidating CGMap's functionalities, Cope and Kalantzis (2023c) emphasize its integration with Open AI's Chat GPT for practical purposes. This integration serves to recalibrate the foundation Large Language Model (LLM) through four distinct mechanisms:

1. Prompt engineering, which refines the queries posed to the LLM by the chatbot.
2. Fine-tuning, involving the supplementation of the LLM with trusted textual data.
3. Ontology supplements, incorporating human-crafted discipline and domain understandings as expressed in data models and interoperability standards.
4. Human moderation, ensuring transparency in interactions with the LLM and mandating human oversight of every AI response.

Additionally, Cope and Kalantzis (2023c) explain that in CGMap, the AI conducts iterative analyses of student work, with each pass focusing on a key aspect of the genre drawn from a rubric. Feedback and suggestions are provided for each aspect, allowing students to run multiple reviews to explore various responses. Importantly, in future implementation the students will retain the ability to query the AI response.

OpenAI's ChatGPT-3.5 via an application programming interface (API) that offers AI feedback to learners on their generated multimodal artifacts. The prompt served to the GPT consists of the instructions to summarize the student text, system instructions, and a request to review the text based on one of the review criteria (Tzirides et al., 2023a) aligned with the assessment rubric (see Figure 2) used in blind peer-review. The shared assessment rubric is grounded in the multiliteracies framework Learning by Design (New London Group, 1996) (see Figure 2) and students were required to incorporate revisions based on the feedback received in each iteration, culminating in a final submission. It is important to note that the design of the prompts was intentionally verbose, mirroring prompts used for peers, to ensure transparency despite their wordiness. This verbosity aimed to scrutinize higher-level "knowledge processes" rather than focusing solely on specific facts. Such an approach sought to address acknowledged shortcomings, such as hallucinations and sourcing issues, as explicated by Cope and Kalantzis (2023c).



Knowledge Processes	Annotation codes	Review Criteria	Marker Nouns (or their equivalent verbs)	0	1	2	3	4
EXPERIENCING Experience	EXP+ EXP- Personal or professional reflection	Clarity and effectiveness of this work as reflected in the connections between this work and the creator's own interests, motivation and experience. Reviewers: What more would you like to know about the creator's interest and motivation to explore this field or theory? Connect to your own experience—in what ways might the chosen area of inquiry be of interest or relevance?	experience, interest, motivation, background, belief, understanding, opinion, perspective	Experiential motivation is not explained.	Experiential basis is explained in limited way.	Good explanation of the experiential basis.	Experiential basis well explained.	Experiential basis explained comprehen-sively.
Evidence	EVI+ EVI- Empirical evidence	Effectiveness in addressing practical questions, sourceable information or citable data from empirical research in order to indicate the significance of this area of activity or inquiry. Reviewers: What other empirical material would you like to see? Do you have suggestions for additional research data or informational source material?	evidence, data, fact, demonstration, identification, observati exploratio documen source, in	Empirical basis not provided.	Empirical basis provided in limited way.	Good explanation of the empirical	Empirical basis well explained.	Empirical basis explained comprehen-sively.
CONCEPTUALIZING Concepts	NAM+ NAM- Concept naming.	Appropriateness and range of concepts to the case being made. Clear definitions. Reviewers: Suggest other concepts that may be required, and where or how definitions of concepts might be more clearly defined.	concept, defiinition categoriz classifica					
Theory	THE+ THE- Theory	Conceptual connections and coherence as a model of the world, and clarity of links between related concepts or important distinctions. Range of key theories and theorists in agreement or in disagreement. Reviewers: Suggest connections that might be made between concepts so the theory is clearer. Suggest other angles or theoretical perspectives that may be relevant.	Theory, hypothes propositi principle					
ANALYZING Reasoning	REA+ REA- Explanation and reasoning.	Effectiveness of the explanations of field, theory or practice under consideration. Reviewers: How sound is the reasoning? Suggest ways in which the reasoning could be more powerful and the explanations clearer.	reason, function, calculatio investigat substanti justificati logic, pat					
Critique	CRI+ CRI- Critical analysis	Awareness of critiques of the theory or practice, and the limits of its scope and applicability. Reviewers: What is the level of understanding of the limits of the creator's own choices? What other lines of critique spring to mind for you? Are there alternative, competing or conflicting theories or	criticism, misunder overlooki neglected disagreem refutatio					
APPLYING Application	APP+ APP- Appropriate application	Explanation of the ways in which ideas presented might translate into practice. Defense of the examples of application provided, gaps in knowledge, and potential for further application and possible measures of effectiveness. Reviewers: What else would you like to know about implementation and effectiveness?	applicati implemen practice, solution, verificati					
Innovation	CRE+ CRE- Creative and innovative application	Actual or possible applications in different contexts that are innovative, or which demonstrate creative thinking or practice. Reviewers: Suggest gaps, innovative or creative potentials, such as lateral or hybrid applications, whether realistic or exciting but perhaps far-fetched possibilities.	innovatio transform change, inventive originality					possibilities explored.
PRESENTATION Communication	COM+ COM- Clarity of communic-	Clear communication of the case, structuring of the work (for instance, using different heading levels in Scholar's structure tool). The quality, range and relevance of embedded media used in the presentation. Textual coherence, connecting media into the		Poor presentation, unclear structure, little	Presentation, structure and media needs improvement.	Good presentation, structure and use of media.	Very Good presentation, structure and use of media.	Excellent presentation, structure and use of media.

Figure 2: Assessment Rubric based on "Learning by Design" Knowledge Processes

The workflow of the main project is depicted in Figure 3. The project begins in Week 1 of the course and concludes in Week 6/8 when final works are published and shared within the course community as well as added to the student's personal portfolio.

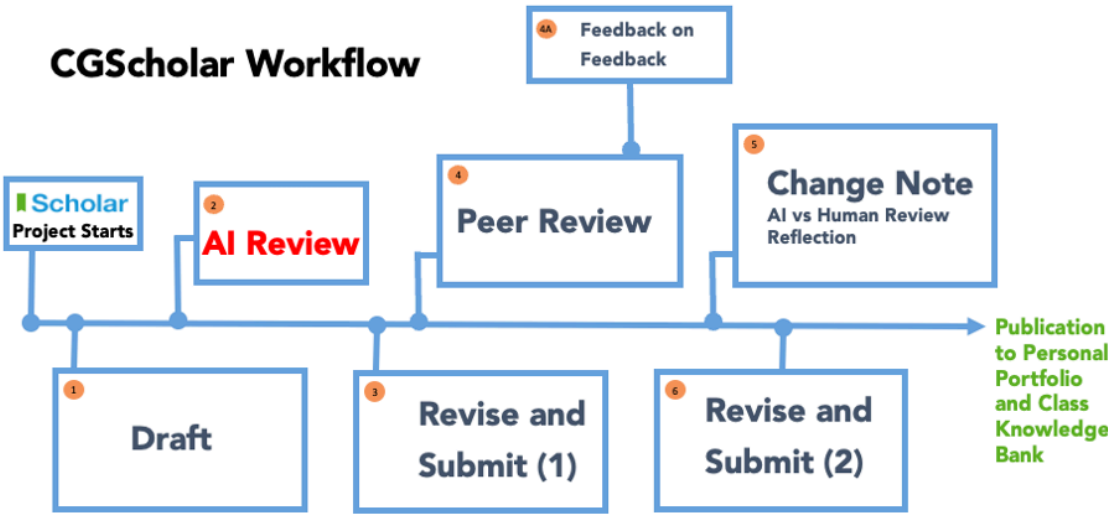


Figure 3: CGScholar Project workflow with AI and peer reviews.

As indicated in Figure 3, upon having the complete draft, the participants generated an AI review through CGMap leveraging the Open-AIs ChatGPT-3.5 (see Figure 4). Students then revised it based on the AI reviewer's feedback (see Figure 4) before submitting it for blind peer reviews.

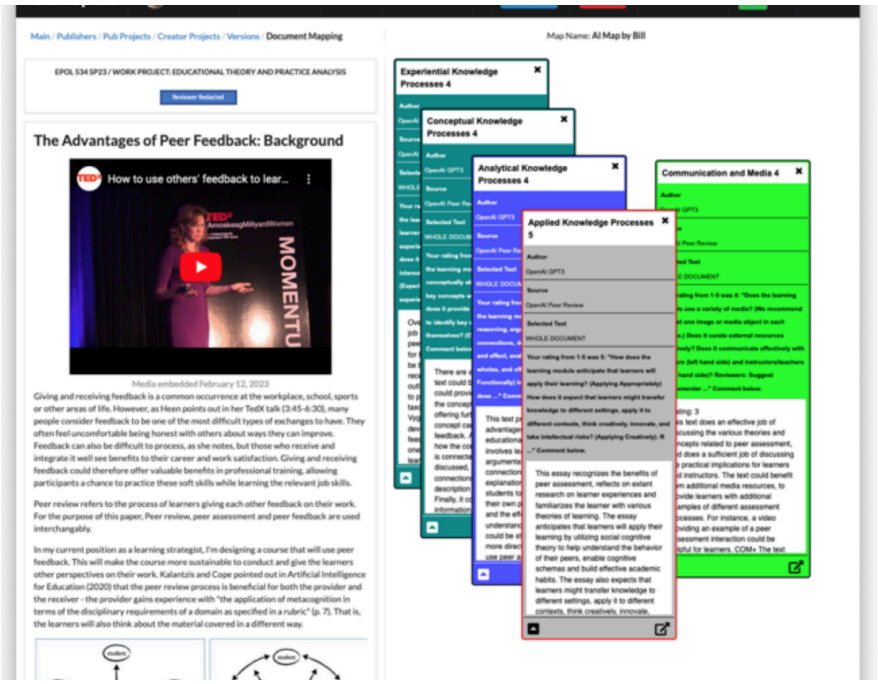


Figure 4: AI review using CGMap for one of the participants

Next, participants engaged in peer review by evaluating two other works using the same rubric criteria within CGMap, which functions similarly to sticky notes on the right side of the screen, linked to highlighted text on the left (see Figure 5).

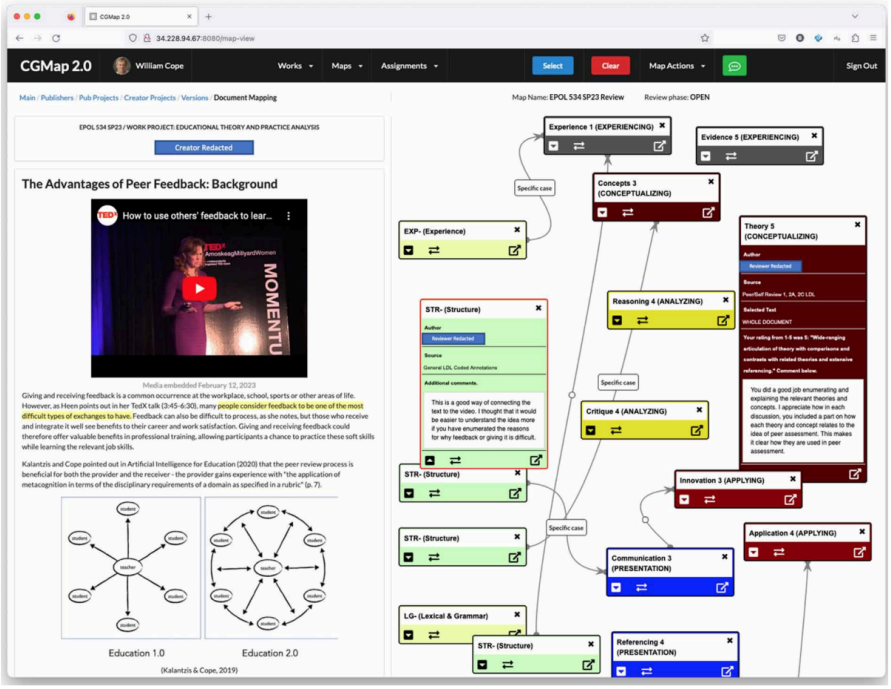


Figure 5: Peer review using CGMap for one of the participants

Later, participants received feedback on their reviews and revised their texts accordingly. Finally, participants compared the feedback from both peers and the AI before making final revisions and proceeding with the review and publication of their work within the course community and personal portfolios (Cope & Kalantzis, 2023d; Tzirides et al., 2023b).
The subsequent post-course surveys was designed to glean insights into students' experiences with both forms of review. Particularly, the investigation is focused on an intervention leveraging a large language model for dialogue applications that was designed to provide graduate students in the programs of focus with AI feedback in addition to anonymized peer reviews (Cho & Cho, 2011) on extended multimodal course work.

Study Design

This exploratory research adopts a convergent mixed-methods approach to investigate the potential of Gen AI for feedback. The study integrates quantitative and qualitative data obtained through a survey to provide a deeper understanding of the involved feedback processes.

Research Questions

The following research questions aim to comprehensively investigate the perceptions and experiences of university students regarding feedback provided by both peers and AI:

1. How are feelings toward AI related to the perceived Quality, Usefulness, and Actionability of AI feedback?
    - Null Hypothesis (H0): There is no significant relationship between feelings towards AI and the perceived Quality, Usefulness, and Actionability of AI feedback.

    - Alternative Hypothesis (H1): There is a significant relationship between feelings towards AI and the perceived Quality, Usefulness, and Actionability of AI feedback.
2. How do participants' perceptions of the Quality, Usefulness, and Actionability of AI and peer reviews differ?
    - Null Hypothesis (H0): There is no significant difference in participants' perceptions of the Quality, Usefulness, and Actionability of respective AI and peer reviews.
    - Alternative Hypothesis (H1): There is a significant difference in participants' perceptions of the Quality, Usefulness, and Actionability of respective AI and peer reviews.
3. How are individual characteristics (age, gender, and education) associated with the perceptions of AI and peer reviews in terms of Quality, Usefulness, and Actionability?
4. What are the perceived benefits and drawbacks of peer reviews and AI reviews as reported by students following their experience with both feedback modalities? In what ways can peer reviews and AI reviews complement each other in the feedback process?

Participants

The participant pool for this research study encompassed a total of 295 individuals. Among them, 57 participants were drawn from the Spring 2023 semester, 143 came from the Summer 2023 semester while the remaining 95 came from the Fall 2023 semester. These individuals of mixed genders and ethnic/racial backgrounds, aged 18 years and above, were enrolled in Non-degree, Master's, Doctorate, and Other degree programs within the College of Education. All participants, along with their academic pursuits, maintained professional careers as educators coming from a wide spectrum of academic disciplines. The recruitment of participants for this research endeavor was executed through course announcements in the academic year programs.

Data Collection

The data for this study was collected through a structured survey administered to participants who were enrolled in fourteen online higher education courses in the College of Education at a US Mid-Western University during the academic term of Spring, Summer, and Fall 2023. For this study, we collected information by asking students questions after their course ended using post-course surveys. We gave them surveys to find out what they thought about peer and AI reviews that were used during the course. We focused on 5-point Likert-scale and open-ended questions where students could write about their experience of peer and AI reviews respectively (check Appendix 2 for the survey questions).

Ethical Considerations

Ethical principles, including informed consent and data privacy, were strictly adhered to throughout the data collection process (see Appendix 1 for the IRB document). Participants were informed about the purpose of the survey and the voluntary nature of their participation. It

is noteworthy to mention that the students who participated in the survey did so voluntarily, without any compulsion or influence on their course scores or grades.

Instruments

The primary instrument utilized in this research study was multiple post-course surveys administered voluntarily each term through the Qualtrics survey platform. The survey aimed to gather valuable insights from participants at the conclusion of the program, eliciting their reflections, experiences, and perceptions related to various aspects of the course. Using a Likert-type scale (0-5), participants expressed nuanced opinions on respective feedback. Coupled with open-ended questions that encouraged detailed qualitative insights, fostering comments, reflections, and suggestions to enrich the survey data.

Data Analysis

Methods

Quantitative data were subjected to statistical analysis using STATA software, employing methods such as mean calculations, paired samples t-tests, and regression analysis to explore patterns and relationships within the dataset. First, we destring the non-numeric variables for mathematical operations, which led to the creation of several dummy variables. We employed regression analysis to get a straightforward and richer measure of linear relationships between variables which are simpler to interpret as compared to the typical correlation. Furthermore, the response data was analyzed using a paired sample t-test to assess the statistical significance of the two sets of survey responses. Similarly, descriptive and summary statistics were used to provide an overview of some of the analyzed variables.

Qualitative data obtained from the open-ended questions were analyzed using thematic analysis to examine the opinions of students regarding the advantages and disadvantages of both peer and AI review processes, as obtained from post-course surveys. The study's methodology, utilized Ryan and Bernard's (2003) thematic analysis framework, leveraging specific scrutiny techniques including repetitions, similarities, and differences; and process techniques such as cutting and sorting, and word co-occurrence. Data from the surveys were compiled into a Microsoft Excel spreadsheet, with each participant's response allocated to a respective sheet alongside the corresponding review type (peer or AI) and each review type’s advantages and disadvantages as perceived by the participants based on their experience. Next, we began the coding process by familiarizing ourselves with the data. This involved reading through all the responses to gain an understanding of the content. Through a systematic approach, we identified recurring themes and patterns in the responses by looking for repetitions, similarities, and differences in the participants' comments. Themes were coded to represent participant ideas using an inductive process (Ryan & Bernard, 2003), separately applied to four sheets covering peer and AI review advantages and disadvantages respectively. The coding process involved familiarization with the data, followed by the systematic application of codes to categorize responses accordingly in Excel. Findings from the thematic analysis were reported descriptively, with emphasis on key themes (based on the number of occurrences) that were identified, supported by quotes from participants, providing a comprehensive understanding of students' perceptions towards peer and AI review processes.

Descriptive Statistics and Visualizations

The age distribution within the sample is diverse, primarily concentrated in the 25-35 years old category (43.39%), followed by 35-45 years old (28.14%) and 45+ years old (22.03%) groups, reflecting a mature and varied age range. A smaller fraction (5.42%) falls into the 18-24-year-old bracket, offering a nuanced age representation.

Regarding gender, the majority of participants identify as female (68.47%), followed by males (28.47%), non-binary (1.02%), and a minimal percentage represents 'Other' genders (0.68%). While some participants chose not to disclose their gender (1.36%).

Educationally, participants indicated a high level of attainment, with a significant proportion pursuing Master's degrees (58.64%), while others pursuing Doctorates (38.98%). Smaller percentages are working toward Certificates (1.02%) or non-degree qualifications (0.68%), with a similar percentage reporting 'Other' educational backgrounds (0.68%).

Characteristics of the sample (N=295)	
Age, %	
18-24 years old	5.42
25-35 years old	43.39
35-45 years old	28.14
45+ years old	22.03
Prefer not to say	1.02
Gender, %	
Female	68.47
Male	28.47
Non-binary	1.02
Other	0.68
Prefer not to say	1.36
Education, %	
Master's	58.64
Doctorate	38.98
Certificate	1.02
Non-degree	0.68
Other	0.68
Location, %	
Africa	0.68
Asia	9.49
Australia	0.34
Europe	1.02
North America	85.75
South America	1.69
Other	1.02

Table 1: Participant Characteristics

Geographically, the majority of participants originate from North America (85.75%), with Asia representing a smaller but notable percentage (9.49%). Minimal representation is observed from Africa, Australia, Europe, South America, and 'Other' locations.

Moreover, as illustrated in Figure 6, the distribution of attitudes toward AI reveals that a predominant 57% express being curious regarding AI's role in reviewing and providing feedback on their work followed by 22% who expressed excitement.

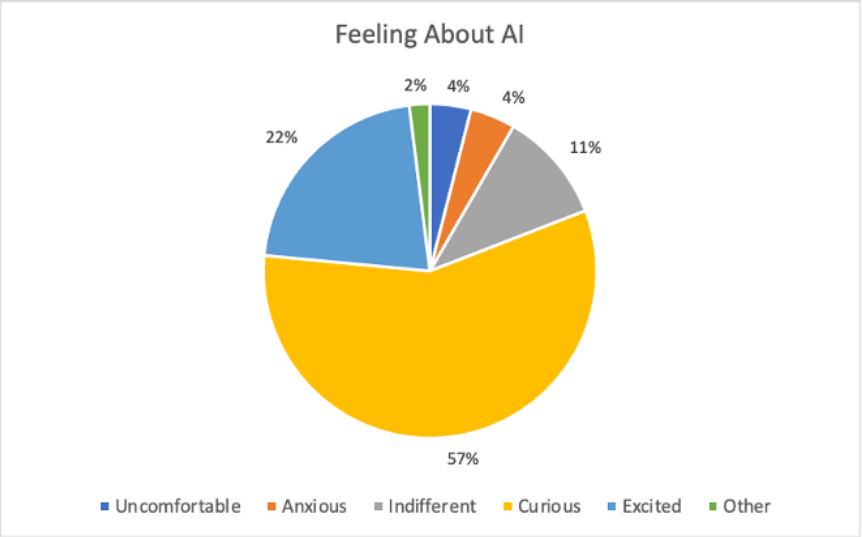


Figure 6: Feeling about AI reviewing work.

In addition, Figure 7 delineates that 41% of participants exhibit a preference for both AI and peer reviews, whereas 35% leaned for peer reviews exclusively.

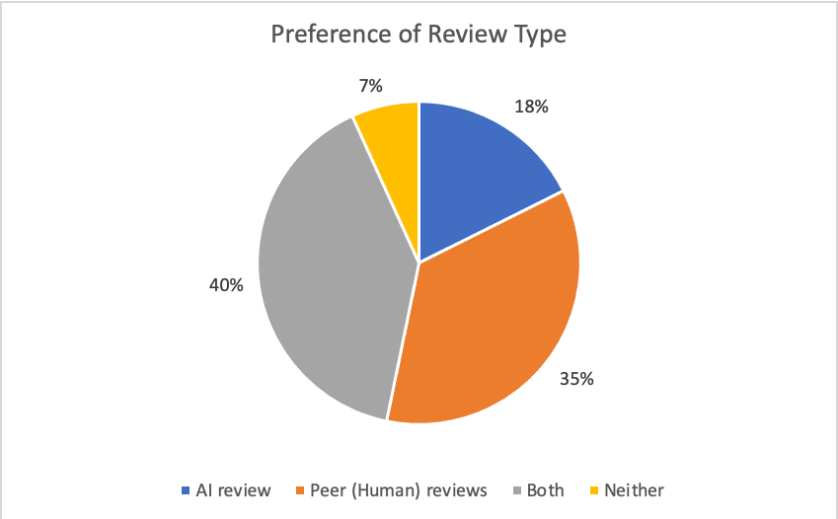


Figure 7: Preferred review type

Results and Findings

RQ1: What is the relationship between feelings towards AI and the perceived Quality, Usefulness, and Actionability of AI feedback?

The analysis of participants' sentiments regarding AI review: those feeling anxious and uncomfortable and those feeling excited and curious showcased a statistically significant relationship between these two. This negative coefficient (-0.377) suggests an inverse relationship, as feelings of excitement and curiosity increase, feelings of anxiety and discomfort decrease. The moderate *R-squared* value implies that these two variables are associated but do not entirely account for each other. The ANOVA results further confirm that these emotional responses are interconnected and contribute significantly to participants' perceptions of AI feedback. The *F-statistic*, a measure of this difference, is 140.67 with a *p*-value less than 0.0001. This low p-value suggests that the observed difference is unlikely due to random chance.

Perceived Quality - The analysis reveals a compelling relationship between participants' sentiments and the perceived quality of AI review. For those feeling curious or excited, the study indicates a significant positive association, supported by a statistically significant coefficient of 0.57 (*t-statistic* = 5.08, $p < 0.05$). This suggests that, on average, individuals with positive emotions rated AI review 0.57 units higher than those feeling anxious or uncomfortable. Conversely, participants feeling anxious or uncomfortable show a significant negative association with AI review quality, evidenced by a statistically significant coefficient of -0.59 (*t-statistic* = -3.38, $p < 0.05$). This implies that, on average, individuals with negative emotions rated AI reviews 0.59 units lower than those feeling curious or excited. These results underscore the influence of participants' emotional states on their evaluations of AI review

quality. Notably, these differences are considerable considering the responses were measured on a 5-point Likert scale. Participants who reported higher levels of curiosity and excitement towards AI were more likely to perceive AI reviews as having higher quality while participants who reported higher levels of anxious and uncomfortable towards AI were more likely to perceive AI reviews as having lower quality.

Perceived Usefulness - The analysis of the relationship between the perceived usefulness of AI review and participants' emotional states yields meaningful insights. For participants feeling curious or excited, there is a significant positive association (*t-statistic* = 4.71, $p < 0.05$) with a coefficient of 0.65. This implies that individuals with positive emotions rated the usefulness of AI review 0.65 units higher, on average, than those who reported feelings of anxiety or discomfort. Conversely, for participants feeling anxious or uncomfortable, there is a significant negative association (*t-statistic* = -3.96, $p < 0.05$) with a coefficient of -0.83. This indicates that, on average, individuals with negative emotions rated the usefulness of AI review 0.83 units lower compared to those who reported feelings of curiosity or excitement. These differences are particularly substantial considering the responses were measured on a 5-point Likert scale.

Perceived Actionability - The analysis of the relationship between the perceived actionability of AI review and participants' emotional states reveals substantial findings. For participants feeling curious or excited, there is a significant positive association (*t-statistic* = 6.06, $p < 0.05$) with a coefficient of 0.85. This implies that individuals with positive emotions rated the actionability of AI review 0.85 units higher, on average, than those who reported feelings of anxiety or discomfort. Conversely, for participants feeling anxious or uncomfortable, there is a significant negative association (*t-statistic* = -6.50, $p < 0.05$) with a coefficient of -1.36. This indicates that, on average, individuals with negative emotions rated the actionability of AI review 1.36 units lower compared to those who reported feelings of curiosity or excitement. Similar to the previous two, these differences are substantial, especially considering that the responses were gauged on a 5-point Likert scale.

Overall, the results reject the null hypothesis (H0), indicating a significant relationship between feelings toward AI and the perceived Quality, Usefulness, and Actionability of AI feedback. We found that positive emotions, like curiosity and excitement, significantly boost the perceived quality, usefulness, and actionability of AI-assisted feedback. Conversely, those feeling anxious or uncomfortable displayed a substantial negative association with quality, usefulness, and actionability. These findings underscore the powerful influence of emotional states on participants' evaluations of AI feedback in terms of its quality, usefulness, and actionability, providing valuable insights for understanding user experiences in AI-assisted feedback scenarios. However, it is crucial to acknowledge the potential interdependence between feelings toward AI and the perceived quality, usefulness, and actionability of its feedback. Participants' positive or negative experiences during reviews may significantly shape their overall sentiments towards AI. Recognizing and disentangling these influences is vital for a nuanced understanding of how emotional responses may impact the objective evaluation of AI feedback, emphasizing the importance of more careful examination in research and practical applications.

RQ2: How do participants' perceptions of the Quality, Usefulness, and Actionability of AI and peer reviews differ?

Table 2 below showcases the output from a paired t-test (for more detail, refer to Appendix 4) comparing the mean quality, usefulness, and actionability ratings given by students for AI and peer review. Correspondingly, for all three dimensions, we reject the null hypothesis (H0), that the mean difference between AI and peer review ratings is zero.

Variables		**Mean**	**Std. dev.**	**Min**	**Max**	***p* value**	***t value***
Quality	AI	2.58	0.81	0	4	$p < 0.05$	-3.11
	Human	2.80	0.93	0	4		
Usefulness	AI	2.20	0.99	0	4	$p < 0.05$	-4.98
	Human	2.61	1.05	0	4		
Actionability	AI	2.69	1.03	0	4	$p < 0.05$	-3.40
	Human	2.93	0.89	0	4		

Table 2: Statistical Insights into Participants' Perception of Quality, Usefulness, and Actionability Ratings for AI and Peer Review
[Scale -
Quality Rating: 0= Very Poor, 1=Poor, 2=Average, 3=Good, 4=Excellent
Usefulness Rating: 0= Not at all useful, 1=Slightly useful, 2=Moderately useful, 3=Very useful, 4=Excellently useful
Actionability Rating: 0= Very difficult, 1=Somewhat difficult, 2=Neither easy nor difficult, 3=Somewhat easy, 4=Very easy]

Quality of Reviews
The paired t-test reveals a statistically significant difference in the mean quality ratings between AI and peer reviews ($t = -3.11$, $p < 0.05$). Participants, on average, rated peer reviews higher by approximately 0.21 units compared to AI reviews.

Usefulness of Reviews
Similarly, the paired t-test indicates a significant difference in the mean usefulness ratings between AI and peer reviews ($t = -4.98$, $p < 0.05$). Participants, on average, found peer reviews more useful by approximately 0.40 units compared to AI reviews.

Actionability of Reviews
The paired t-test demonstrates a significant difference in the mean actionability ratings between AI and peer reviews ($t = -3.40$, $p < 0.05$). Participants, on average, rated peer reviews as more actionable by approximately 0.24 units compared to AI reviews.

The evidence suggests that, on average, peer review ratings are significantly higher than AI review ratings for quality, usefulness, and actionability. These differences are substantial, especially considering that the responses were gauged on a 5-point Likert scale. So, in

layman's terms, there is a statistically significant difference between the ratings for the three chosen dimensions given by humans and the AI system, and the human ratings tend to be perceived more favorably.

RQ3: How are individual characteristics (age, gender, and education) associated with the perceptions of AI and peer reviews in terms of Quality, Usefulness, and Actionability?

The findings present nuanced insights into the relationship between age and the perceived quality, usefulness, and actionability of AI and human feedback (for more detail, refer to Appendix 5). Participants aged 45+ consistently rate both AI and human feedback the highest in quality, with AI scoring a mean of 2.70 and peers 2.97. In contrast, the 18-24 age group tends to give lower ratings, favoring peers (Mean = 2.69) over AI (Mean = 2.13). This pattern persists in the 25-35 and 35-45 age groups. Regarding perceived usefulness, participants aged 25-35 rate AI the lowest (Mean = 2.13), while the 35-45 and 45+ groups give higher ratings (Mean = 2.24 and 2.35, respectively). Human feedback consistently receives higher ratings across all age groups, peaking at 2.78 for the 45+ group. In terms of actionability, the 25-35 age group exhibits the highest confidence in AI (Mean = 2.83), contrasting with the 18-24 group, which shows a lower preference for AI (Mean = 2.38). For human feedback, the 25-35 group rated it the highest (Mean = 3.07) and the 18-24 age group had the lowest (Mean = 2.69).

In gender-related perceptions of AI and human feedback, females consistently rate both higher than males. For quality, females score peer reviews at 2.82 and AI at 2.64, while males rate peer reviews lower (Mean = 2.80) and AI even lower (Mean = 2.5). Regarding usefulness, females rate AI at 2.82 and peer reviews at 2.65, while males rate AI lower (Mean = 2.05) and peer reviews slightly lower (Mean = 2.57) as well. In terms of actionability, females rate AI at 2.63 and peer reviews at 2.98, whereas males rate AI and peer reviews slightly higher (both with Mean = 2.80).

In education, individuals pursuing a Doctorate degree also give favorable ratings on perceived quality, with AI reviews (Mean = 2.69) slightly below peer reviews (Mean = 2.79). Participants enrolled in Master's degree provided high ratings for both AI (Mean = 2.52) and human (Mean = 2.79) reviews as well on perceived quality. For perceived usefulness, individuals enrolled in a Doctorate degree, however, rate AI feedback slightly higher (Mean = 2.35) than Master's degree pursuing participants (Mean = 2.11), and the same was true for the case of peer reviews as well. Regarding perceived actionability, individuals pursuing a Doctorate degree provided the highest rating for AI feedback (Mean = 2.86), while those enrolled in a Master's degree rated it lower (Mean = 2.59). For human feedback, those pursuing a Doctorate degree are again associated with a higher rating (Mean = 2.95) than a Master's degree (Mean = 2.91). Participants enrolled in non-degree and certificates present diverse patterns, indicating a less predictable relationship between education and perceived quality, usefulness, and actionability.

These findings underscore the influence of age, gender, and educational background on perceptions of AI and human feedback, highlighting the necessity for customized feedback mechanisms tailored to specific demographic groups. Although no conclusive overall pattern emerged, these variations emphasize the significance of demographic factors in comprehending feedback perceptions.

RQ4: What are the perceived benefits and drawbacks of peer reviews and AI reviews as reported by students following their experience with both feedback modalities? In what ways can peer reviews and AI reviews complement each other in the feedback process?

The following findings outline the identified advantages and disadvantages of human/peer and AI reviews based on this study implementation. The coded themes in each section capture the identified benefits and drawbacks followed by a detailed examination of the specific outcomes associated with the respective review type. This exploration sheds light on the interplay between AI and human expertise in the context of educational assessment.

Advantages and Disadvantages of Peer Review

This sub-section covers the identified themes from the thematic analysis of the pros and cons of utilizing peers for providing feedback and evaluation in educational settings (in order of the number of occurrences), drawing on data from the post-course survey student responses.

Advantages

Students listed several favorable aspects of peer review. Below are the coded themes that encapsulate the identified advantages of peer review:

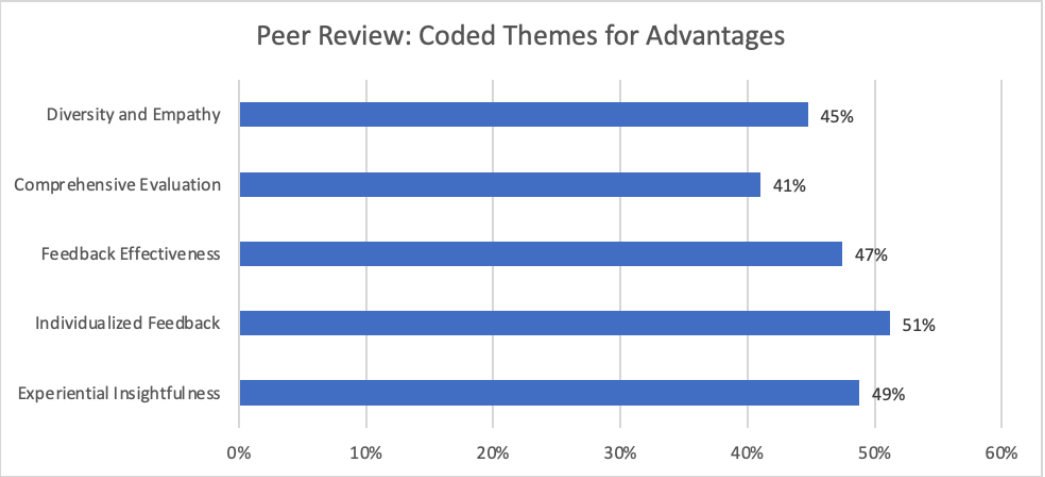


Figure 8: Coded Themes for Identified Advantages of Peer Review

Students seemed to highly appreciate the personalized nature of the feedback received from their peers, recognizing its specificity and relevance to their individual assignments. By tailoring feedback to the context of each student's assignment, peers appear to have offered valuable insights that were directly relevant to the student's learning goals and objectives. This is clearly seen in the following quotes:

> "[The peer reviews was] on point and specific, suggests recommendations that I hadn't thought of." (Participant #7)

> "The peer reviews provided me with very specific feedback not only about ways that I could beef up my paper but also specific mechanical errors that needed fixed." (Participant #30)

> “The review is very thorough, the feedback is more personal and from multiple people - you get multiple opinions.” (Participant #231)

> “Human feedback provides a more individualized and specific response…can be very beneficial for improvement.” (Participant #261)

Students highlighted the expertise and experience peers brought to the feedback process as particularly beneficial. Students valued the depth of insight provided by peers who demonstrated a strong understanding of the subject matter:

> “Peers develop their own interpretation and perspective on the work. This provides better feedback and understanding of how well you are communicating your ideas. It also gives you more information and other perspectives to incorporate into your work.” (Participant #97)

> “The peer reviews were able to take from their experience and their knowledge and apply that into their suggestions for my work. The suggestions felt more impactful as I appreciated the time and effort it took to look over my work.” (Participant #190)

> “[Peers can] be specific and understand the larger context of the work, they brought individual expertise and experiences.” (Participant #220)

> “[Peers] have their own perspectives and experiences that they are able to pull ideas and suggestions from. Humans have more ability to review and think and process ideas overtime and from different perspectives.” (Participant #235)

The effectiveness of feedback in constructively guiding students' progress and addressing course requirements was also underscored. Students appreciated feedback that not only identified areas for improvement but also offered actionable suggestions tailored to their assignments. Several students vocalized their thoughts consistent with the overarching theme, mentioning that:

> “Peer reviewers understood the assignment which for this class was quite challenging. Peer reviewers were also educators in some capacity, so they understood the professional context of the work.” (Participant #43)

> “My [peer] reviews were decent, and gave direction on where to improve or add detail. The best suggestions were actionable.” (Participant #48)

> “Peer reviews are great in that one gets constructive criticism and useful suggestions.” (Participant #101)

> “[Peer review was] encouraging and yet with actionable, constructive feedback.” (Participant #283)

Moreover, the diverse perspectives and empathetic approach evident in peer feedback were valued for creating an inclusive and supportive learning environment. Students acknowledged

the importance of feedback that considered various unique viewpoints and demonstrated emotional intelligence:

> “Peers provide feedback with very specific details. Human reviewers offered multiple viewpoints that praised or critiqued different parts of the work, that helped create a full picture.” (Participant #14)

> “Humans can review your work through a peers perspective. Humans can read emotion.” (Participant #83)

> “The human reviews had more emotion and a sense of encouragement in them. The human reviews could relate to the content and process of the project.” (Participant #94)

> “[Peers were] encouraging, [and had] emotional resonance.” (Participant #196)

Likewise, peer feedback processes were praised for their efficiency and thoroughness in providing detailed analysis of students' work. Students found value in the nuanced feedback they received, which facilitated their learning and growth with many expressing as follows:

> “Human reviewers are often able to provide more in-depth and nuanced analysis of papers than AI systems. People can identify subtle flaws and provide feedback on how to improve the paper/work project.” (Participant #21)

> “[Peers] shared resources and provide in-depth comments to revise my project. I get to learn [about] their projects because they often refer to their project in explaining their comments.” (Participant #28)

> “It was great to know that they picked up on things the AI did not and they understanded nuanced or difficulty subjects. The provided good suggestions for adding interactivity work.” (Participant #136)

> “Peer reviews offer a deeper understanding of the context and subject matter, allowing for more nuanced and relevant feedback tailored to the specifics of the assignment.” (Participant #194)

In addition to the mentioned themes, the following positive connotations were observed: ‘honesty’, ‘meaningful’, ‘discourse’, ‘relatability’, ‘validation’, and ‘innovation’. It is important to note that 27 out of 295 participants either responded with not applicable (N/A), did not receive reviews for their submissions, or received it very late for it to be useful to submit a response.

Disadvantages

Students reported several aspects of peer review that they found unfavorable or unhelpful. Presented below are the coded themes representing the identified disadvantages of peer review:

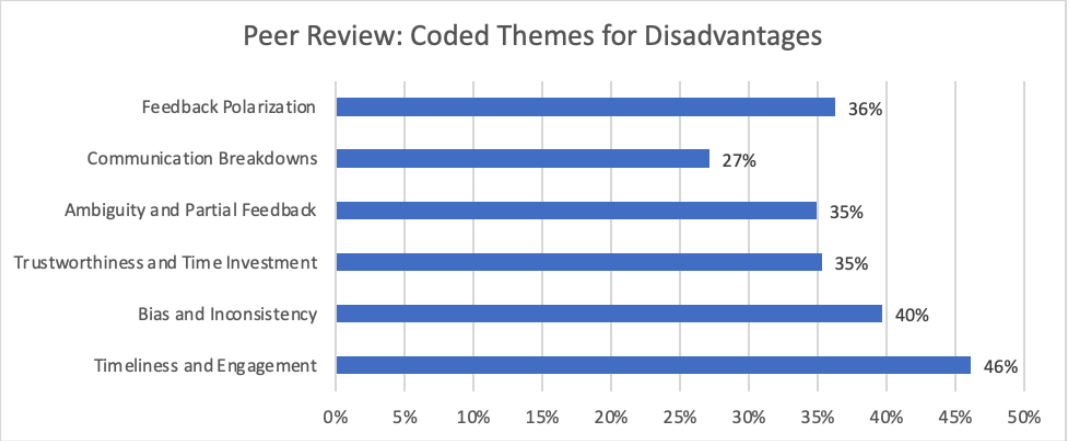


Figure 9: Coded Themes for Identified Disadvantages of Peer Review

Students expressed various concerns regarding the peer-review process as reported in the post-course survey. One recurring issue was the extent of dependency on reviews and discussions for their learning process, along with concerns about engagement and participation. A significant portion of students conveyed opinions congruent with this theme, noting that:

> "Takes time and something do not come through. Very dependent on the experience of the peer." (Participant #9)
>
> "Sometimes you do not receive it fast enough, and sometimes you do not receive it at all." (Participant #64)
>
> "...not being able to converse or explain in real time [with peers]." (Participant #85)
>
> "you are at the discretion of the peer and if they do not submit their part in a timely manner, it may put you behind." (Participant #112)

Another significant concern was the presence of subjective viewpoints or biases influencing feedback, leading to inconsistent opinions and feedback. This issue emerged prominently as the second most frequently mentioned disadvantage with numerous students who expressed this sentiment, stating, that:

> "Some peers are nitpicky and subjective about certain points such as whether or not a media element is effective…" (Participant #13)
>
> "Peer/human reviews involve a subjective assessment of a paper's quality (even when a rubric is provided). People have different opinions on what makes a good paper, and their evaluations can be influenced by personal biases and preferences. Human reviewers may have different levels of experience, expertise, and biases, which can lead to inconsistent evaluations of papers/work projects." (Participant #21)

> “Peers might hold their own biases or understandings of how things should be done, so that might influence their rating of others' work.” (Participant #88)

> “Peer reviews could be biased. Human reviews may be subjective.” (Participant #262)

Students also noted a tendency for feedback to be overly skewed towards positive or negative sentiments, lacking balance, and sometimes overly concentrating on specific aspects while neglecting others. Many students concurred with this theme, and shared that:

> “...sometimes, they [peer reviewers] are worried about upsetting the writer, so reviews can be overly positive and not particularly helpful.” (Participant #22)

> “...I got several low scores combined with positive, general comments and no suggestions, so it was impossible to figure out why the work was scored so low. While the rubric seems clear, it doesn't seem to be applied evenly by different peers.” (Participant #62)

> “[The] disadvantage to the peer reviews was the somewhat condescending tone included in the review.” (Participant #93)

> “I feel that some parts of the reviews were bumped up to a higher score than should have been which could be peers being nice...” (Participant #209)

The quality of feedback and discussions was scrutinized as well, considering the level of expertise and effort behind the feedback provided or discussions held. Several students reiterated sentiments aligned with this theme, underscoring that:

> “Some of these review comments miss the mark because the reviewer seems to have no background knowledge in the topic and may not have read closely to identify certain explanations provided in the text that makes clear the scope of the work.” (Participant #26)

> “I only received one peer review and initially it didn't have annotations. I understand that people are busy, but it's frustrating to spend so much time on peer reviews for my classmates and to receive feedack in return that isn't very constructive.” (Participant #69)

> “Depending on the experience level of the peer reviewer and the time pressure they are under, you may not get as much detailed feedback compared to the AI review, which is fairly comprehensive in all rubric criteria.” (Participant #197)

> “The major disadvantage is that not every one knows how to do peer reviews or even be a reviewer. So the information sometimes is not insightful nor robust. [Also] to do a review well requires a lot of work and I believe not all want to put forth that effort thus resulting in sub-par comments or useful critique.” (Participant #277)

Similarly, students highlighted feedback characterized by vagueness, lack of clarity, and superficiality, which fail to provide in-depth insights or meaningful engagement with the subject matter. A notable group of students resonated with this theme, highlighting that:

> "...I've had a handful of very vague reviews that weren't helpful because they were too general." (Participant #27)
>
> "Sometimes too broad or negative, like putting me 1 without thorough explanation." (Participant #99)
>
> "[Peer review was] not thorough and feel empty." (Participant #182)
>
> "The feedback felt lazy and generic in one case. I would have appreciated a more nuanced feedback." (Participant #275)

Lastly, misunderstandings or misinterpretations during feedback provision, especially concerning alignment with course rubrics, leading to confusion or miscommunication between the parties involved were also flagged. A substantial number of students ecoed this sentiment, stressing that:

> "[Peers] can get caught up in the details and subjective in the interpretation of the rubrics." (Participant #10)
>
> "...a peer human reviewer may misconstrue a primary goal for a work by superimposing an assumed specific goal of their own and making specific suggestions based on their incorrect assumption..." (Participant #61)
>
> "My peers didn't always understand the rubric and took away points when they should have. In turn, those points affect my course grade (even if minimally)." (Participant #80)
>
> "[Peers] didn't relate to the rubric at all times. It may also be biased according to their interpretation of the rubric." (Participant #115)

Other than the mentioned themes, several negative attributes were identified, including descriptors such as 'confusing', 'generic', 'inaccurate', 'repetitive', 'non-applicable', 'short', 'unorganized', and 'grade-opacity'. Besides, a subset of 33 respondents out of 295 indicated 'Not Applicable' (N/A), did not receive reviews for their submissions, received them too late to be useful for providing feedback, did not encounter any disadvantages to report, or expressed satisfaction with the peer-review process.

Advantages and Disadvantages of AI Review

This subsection delves into the identified themes extracted from a thematic analysis concerning the advantages and disadvantages of employing AI for feedback and evaluation in educational contexts. These themes are presented in order of occurrence frequency and are derived from the data obtained from post-course survey responses provided by students.

Advantages
Similar to peer reviews, students identified several favorable aspects of AI review. The ensuing coded themes encapsulate the identified advantages of AI review:

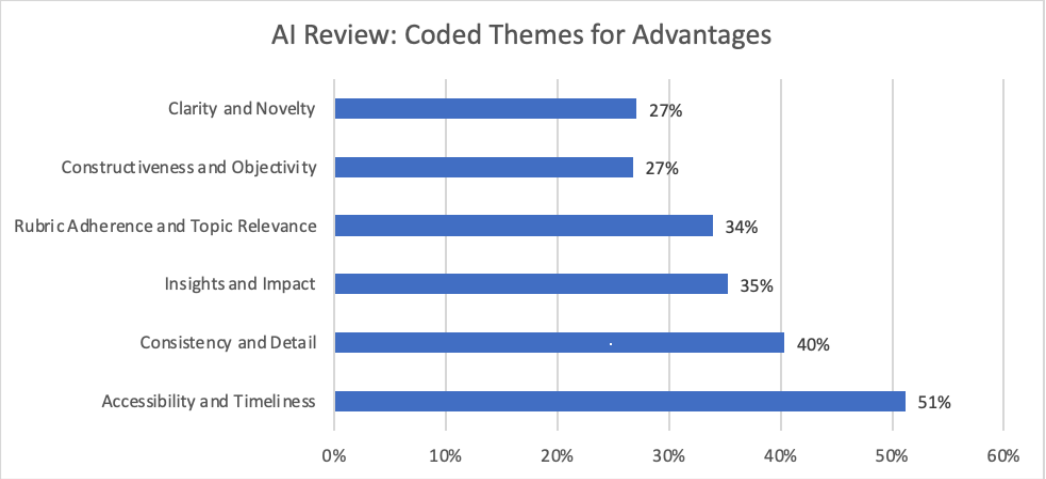


Figure 10: Coded Themes for Identified Advantages of AI Review

Students emphasized the accessibility and immediacy of feedback provided by AI systems as the most significant advantage. The feedback was readily available and delivered promptly, and could be employed multiple times, offering a flexible and accommodating approach to assessment. This aspect resonated strongly with many participants affirming that:

> "Immediate feedback that could be immediately reviewed, considered, and incorporated while waiting on human reviews." (Participant #26)
>
> "It is very quick, [and] can be used again for next version." (Participant #77)
>
> "The feedback was immediate. You are guaranteed to have something to help you make improvements." (Participant #129)
>
> "AI Review is very efficient - you get your feedback within 10 minutes or so. The AI review can be utilized as many times as you want throughout your work process." (Participant #230)

The consistency and thoroughness of feedback from AI systems were similarly praised, particularly in offering detailed insights and identifying grammatical errors, structural flaws, and specific content weaknesses. Among the respondents, there was consensus on this theme:

> "The feedback was thorough and provided both pluses and minuses. More consistent." (Participant #17)
>
> "AI reviews are typically more consistent than human reviews (AI systems can be programmed to apply the same criteria consistently to all papers/work projects)." (Participant #21)

> “It was thorough in it's review of my work. It listed out specific areas that needed attention and even provided suggestions on how to improve. It has access to the rubric so it is excellent at aligning my work with what is required by the rubric.” (Participant #190)

> “[The AI review] offers a consistent and unbiased evaluation, minimizing potential errors. Its prompt feedback allows for quick improvements, ensuring timely submission of polished assignments.” (Participant #194)

Students also mentioned that feedback from AI systems provided valuable insights concerning areas for growth, recommended useful resources, and significantly enhanced performance or understanding. Many students concurred with this theme, stressing that:

> “The AI Review made me aware of at least one area in my paper which could be enhanced for a more powerful presentation by adding more specific research data points…” (Participant #61)

> “...you are guaranteed to have something to help you make improvements.” (Participant #129)

> “... It provides a clear and structured evaluation system, making it easier to assess performance, identify areas for improvement, and set goals for growth...” (Participant #241)

> “[AI review] identified theoretical frameworks that I could use that I don't think a peer reviewer would have. It identified a pattern in my writing that helped me see I needed to provide more examples to illustrate my points.” (Participant #289)

Likewise, AI feedback was noted for its alignment with the rubric and the specific topic, ensuring that comments and suggestions were directly related to the established criteria and subject matter of the assignment or task at hand. A substantial number of students validated this theme, asserting that:

> “AI review gave holistic feedback and suggestions based on the rubric. They can check all the content of scholastic resources and connections between them and my article, which is not applicable for peer review.” (Participant #16)

> “AI was more accurate and aligning with the rubric elements helping me to make sure I could check for the necessary content.” (Participant #94)

> “It tells me what has already been covered in terms of rubric, what's already been written well. It also tells me It tells me what has not already been covered in terms of rubric, what should be included to make the work thicker.” (Participant #188)

> “It helped with grammatical errors and pointed out if we were missing key pieces according to the rubric.” (Participant #271)

Students also lauded AI reviews for its clarity and ability to offer unique perspectives or inspirations that aid further refinement and development of their knowledge and understanding. Several students reiterated sentiments aligned with this theme, underscoring that:

> "Quickly aggregates all of the work, [and provides] interesting and unique perspective." (Participant #19)
>
> "...AI reviews offer the ability to offer many different perspectives rooted in many frameworks." (Participant #112)
>
> "Prompting a new idea to take one section of my work & thorough breakdown of how more critique of my theory is necessary." (Participant #150)
>
> "The AI review gave me feedback on things I hadn't caught or realized needed revision. It also provided me with new ideas of things to include in my project." (Participant #277)

Last but not least, students appreciated the constructive and objective nature of AI feedback as well, which maintained neutrality and impartiality, and promoted fair and effective assessment practices. This aspect resonated strongly with many participants, expressing a viewpoint in line with the prevailing theme, that:

> "..It also offers an impersonal viewpoint without any potential social or emotional connection to the work." (Participant #14)
> "There are no hurt feelings when the report comes from a bot with no feelings. =). I understand the limitations and mistakes the review will make and just take the value given to me." (Participant #59)
>
> "The AI was wholly unbiased and impartial; whereas a human might be tempted to be kind, the AI has no such inclination. :)" (Participant #110)
>
> "[The] AI review for [course name] was helpful and relevant, and comparing this to the peer reviews was useful in providing actionable feedback that I used to improve my work. I appreciated the unbiased, non-student perspective of my writing." (Participant #144)

Beyond the discussed themes, the following positive attributes were identified: 'organized', 'high-standard', 'holistic', 'accurate', 'reliable', 'emotion consideration', 'human review complementarity', and 'exciting'. Out of 295 respondents, 6 indicated N/A, citing difficulties utilizing the review due to glitches or confusion, or having not yet generated their AI review.

Disadvantages

While there are advantages, there are also drawbacks. Students identified various aspects of AI review that they deemed unfavorable or unhelpful. Presented next are the coded themes representing the identified disadvantages of AI review:

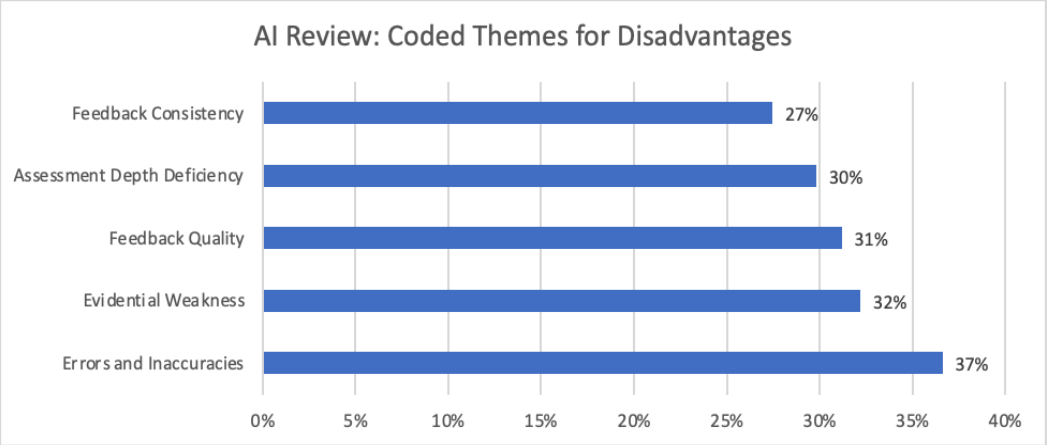


Figure 11: Coded Themes for Identified Disadvantages of AI Review

Students pointed out various issues related to factual errors, inaccuracies, and misinterpretations in AI feedback, indicating a lack of reliability and precision. Many articulated a perspective that resonated with the broader theme, stating that:

> "Sometimes the errors are egregious and whole sections of commentary are useless." (Participant #58)
>
> "[AI reviews showcased] Misinterpretation and innacuracies." (Participant #80)
>
> "Some feedback was not helpful and based on misinterpretations of the text that would be unlikely to happen with feedback from a human reviewer." (Participant #89)
>
> "[AI review] had margin of errors and [provided] feedback to include things that I had included." (Participant #197)
>
> "Sometimes it could give a false score, it doesn't catch typos or grammar mistakes." (Participant #229)

The absence of credible information, specific examples, and contextual understanding to support arguments was highlighted as a significant concern, indicating a weakness in the foundational support of analyses. The resonance of this theme was palpable among participants, as echoed in the comments below:

> "AI cannot understand context, purpose or tone to the extent necessary to make sense of how one human attempts to connect to other humans in writing." (Participant #32)
>
> "Many of the suggestions didn't make sense given the context of the assignment. It also frequently dropped 'intellectual' from 'intellectual disability'. The meaning was changed and AI didn't notice." (Participant #42)

> “Some of the comments were out of context and misdirected by irrelevant text. It didn't add or create new knowledge through an interpretive lens, it just rehashed what was already known.” (Participant #56)

> “[AI review] penaliz[ed] me when it failed to understand the purposeful, specific scope of my work & coming back to the same phrase in its critique.” (Participant #149)

Issues with non-specific, repetitive, or overly generic feedback were identified, pointing to the need for more constructive, detailed, and personalized feedback mechanisms to enhance understanding and improvement. Several respondents offered observations, consistent with the theme, suggesting that:

> “It was repetitive at certain points [and] felt superficial.” (Participant #83)

> “[AI review was] vague, off topic sometimes, no specific suggestions or positive comments, supportive statements or experience sharing. [I] would like AI to provide me with the links and titles, help me with my work deeply.” (Participant #98)

> “No pin points; sometimes too generic of feedback to be meaningful.” (Participant #144)

> “The AI review had a tendency to repeat itself on some of the feedback. It also was a little unclear at times where specifically in my project the AI was suggesting revision for.” (Participant #276)

Students also raised concerns regarding the lack of thoroughness and detailed analysis in AI evaluations, suggesting a surface-level engagement with the material or subject matter. A significant group of students resonated with the holistic theme, detailing that:

> “...the AI review missed some of my smaller points and it was more critical of my project.” (Participant #10)

> “Human reviewers are often able to provide more in-depth and nuanced analysis of papers than AI systems. People can identify subtle flaws and provide feedback on how to improve the paper/work project. AI systems, on the other hand, are limited by their programming and may not be able to provide the same level of detailed analysis.” (Participant #20)

> “[AI review was] not as in-depth as a human review. [It also] doesn't go beyond the rubric.” (Participant #139)

> “The AI review is not as in-depth as a good peer review. At times it does not pick up on structorial issues.” (Participant #176)

Not least of all, students mentioned challenges in maintaining a coherent and supportive feedback system, leading to contradictions and inconsistencies in AI-generated feedback.

The consensus among respondents was evident, with many emphasizing that:

> “The only disadvantage I found was the inconsistent feedback. I reviewed my AI analysis, changed what was in the feedback and scored lower. Then I ran it again without changing anything and the score changed again.” (Participant #119)

> “The feedback sometimes contradicted itself. For example, criterion one gave the two pieces of feedback: ‘The author has specified the range of learning levels for which the learning module might be suitable’ and ‘There is no mention of any other learner groups for which this module might be appropriate.’ Those were in the same section.” (Participant #127)

> “[The] recommendations from AI were difficult to interpret, often contradicting other AI-generated feedback.” (Participant #140)

> “Inconsistent reviews for the same project content, each run producing different scores.” (Participant #178)

Apart from the discussed themes, the following negative connotations were identified: ‘limited interaction’, ‘lacking personalization’, ‘tone’, ‘absent multimedia feedback’, ‘mechanical’, ‘APA and lexical concern’, and ‘long/broad feedback’. Moreover, among the 295 respondents surveyed, 10 indicated 'Not Applicable' (N/A) or cited encountering challenges such as glitches or confusion in utilizing the review system, or they had not yet generated their AI review to report disadvantages. Whereas some respondents (6 out of 295) did not encounter any disadvantages to report or expressed satisfaction with the generated AI reviews.

The qualitative analysis reveals complementary strengths between peer and AI reviews. While participants appear to characterize peer review as exhibiting strengths in personalization, contextual understanding, and diverse perspectives, AI reviews were perceived higher in accessibility, timeliness, detail, objectivity, and consistency. Peer reviews demonstrate higher subjectivity and variability in terms of expertise, clarity, and consistency. Conversely, AI reviews show lower subjectivity but lack the personalization and diverse insights inherent in peer interactions. These findings highlight the complementary nature of peer and AI reviews, suggesting potential benefits in combining both approaches for comprehensive feedback in educational assessments.

Discussions and Reflections

In this study, the first three research questions were approached using quantitative analysis seeking to unravel the connections between emotional states and perceptions of AI feedback (RQ1), discern differences in ratings between peer reviews and AI reviews (RQ2), and understand demographic variations in feedback perceptions (RQ3). These analyses provided valuable insights into the relationships and disparities observed in students' perceptions of quality, usefulness, and actionability of respective review types. The analysis for the first research question revealed that positive emotions correlated with higher evaluations, while negative emotions led to lower assessments. For the second research question, the analysis affirmed that peer reviews consistently received higher ratings than AI reviews across quality,

usefulness, and actionability dimensions (all $p < 0.05$). Regarding the third research question, the analysis revealed nuanced variations in perceptions of AI and human feedback based on demographics. While no definitive pattern emerged, these variations underscore the importance of considering demographic factors in understanding feedback perceptions.

In contrast, qualitative analysis was utilized to explore the advantages and disadvantages of peer and AI reviews (RQ4). Thematic analysis of participant responses revealed nuanced insights into the operational differences involved, and strengths and weaknesses of each review type as well as the potential for integration to enhance both feedback mechanisms in educational settings. Firstly, in examining the perceptions of respective feedback processes, notable differences became apparent:

1. Peer feedback was characterized by delays, contrasting with the immediacy of AI feedback.
2. Peer reviews typically entail a single attempt per student, contrasting with the ability of AI reviews to allow for multiple iterations.
3. Peers often seem to offer feedback in various modes, such as text, audio, pdf, or video, whereas AI feedback primarily consists of one mode (text).
4. Peer reviews are perceived as time-intensive, while AI reviews are perceived as time-efficient.

Secondly, the results of the participants' reflections on peer review revealed that participants seemed to value individualized feedback tailored to students' work, peers' experiential insightfulness and contextual understanding, and facilitation of diverse perspectives. However, challenges such as timeliness, bias, polarization, ambiguity, and inconsistency were noted, signaling the need for improvement in feedback quality and communication. On the other hand, AI reviews seemed to be praised for accessibility and timeliness, clarity, detail, adherence to rubrics, objectivity, consistency, and constructive nature. Yet, concerns were raised about errors, evidential weaknesses, contextual understanding, and feedback quality, indicating areas for refinement in AI systems. Thus, a balanced integration of both approaches could optimize the feedback process, harnessing the strengths of each may result in more comprehensive and quality feedback as well as enhance the overall effectiveness of feedback provided to students.

Pedagogical Implications and Future Directions

Our cyber-social research team has been diligently reimagining and reengineering the utility of Gen AI for feedback, with several recent publications on the topic (Cope & Kalantzis, 2023a, 2024a, 2024b; Tzirides et al., 2023a; Zapata et al., 2023, 2024). Here are five pedagogical implications derived from the findings of this study, relevant to feedback mechanisms in educational settings.

1. *Understanding Emotional Influence*: Educators should recognize the significant impact of students' emotional states on their perceptions of feedback quality. Strategies to recognize and manage (or even foster positive) emotions during feedback sessions or processes may enhance students' engagement and receptiveness to feedback.

2. *Leveraging Peer Reviews*: Despite the prevalence of AI systems, peer reviews remain highly valued for their personalized and diverse perspectives. Educators should

capitalize on the strengths of peer feedback while minimizing the challenges, emphasizing its role in facilitating meaningful learning experiences and actively promoting peer interaction.

3. *Optimizing AI Feedback Systems*: While AI feedback offers advantages in accessibility and timeliness, efforts are needed to address concerns regarding accuracy, evidence-based support, and feedback quality. Further development and refinement of AI systems should prioritize these aspects to enhance their effectiveness in providing targeted and valuable feedback to students with educators embracing an agile pedagogical approach in using these technologies.

4. *Critical Evaluation Skills Development*: To address concerns related to the lack of empirical truth and epistemic provenance in AI-generated content, educators should prioritize the development of critical evaluation skills. At the same time, teaching students to critically evaluate sources, assess the credibility of information, and discern between reliable and unreliable content with a metacognitive reflective lens will empower them to navigate the complexities of AI-generated information effectively, reflecting the suggestions discussed in Cope & Kalantzis (2024a) and Otaki (2023).

5. *Enhanced Feedback Mechanisms:* A balanced integration of both peer and AI review approaches holds promise for optimizing the feedback process. By harnessing the strengths of each approach, educators can create a more comprehensive feedback environment that addresses the diverse needs and preferences of students.

As we look towards the future, the evolution of AI systems is progressing rapidly, driven by innovative techniques such as fine-tuning and advancements like Retrieval-Augmented Generation (RAG) (Balaguer et al., 2024). Fine-tuning involves further training pre-existing models on domain-specific data to enhance their performance on particular tasks, while RAG seamlessly integrates generation with retrieval mechanisms to produce outputs drawing upon external knowledge sources (Balaguer et al., 2024; Ovadia et al., 2023). These ongoing advancements are transforming AI outputs by enabling rapid adaptation to new tasks, enhancing context understanding, and improving overall performance. Recent research has demonstrated the superiority of RAG over traditional approaches, both in terms of leveraging existing knowledge encountered during training and adapting to entirely new knowledge contexts (Ovadia et al., 2023). These findings corroborate the conclusions drawn by Rangan and Yin (2024), who underscore the effectiveness of RAG in improving accuracy, particularly in the context of large-scale models. Additionally, the relatively low initial cost of creating embeddings, which are data representations, further enhances RAG's attractiveness as a viable option for enhancing AI systems (Balaguer et al., 2024).

Our research team, inspired by the promising outcomes, embarked on experiments with RAG to enhance the output generated by AI systems. Using the same research design and data collection methodology, we conducted a post-course survey as part of our Spring 2024 POT A implementation, involving a sample of 45 students. During this survey, students were asked to rate the quality, usefulness, and actionability among other characteristics of peer and AI-generated reviews. Notably, for this iteration, we integrated RAG techniques into the existing LLM utilized for generating AI reviews. Initial observations from the survey data suggest a notable difference in student perceptions between AI-generated and human-generated

reviews, with their preference tilting towards AI-generated reviews. Students seem to appreciate and value the new AI-generated reviews, with many expressing:

> “I was very impressed with the AI review I received. It was far superior to any prior AI reviews I have completed in past courses. The AI review made sense, was well-organized, articulated clear strengths and weaknesses, and made appropriate, useful suggestions for revision.” (Anonymous #3)

> “This is my third class using this system, and the reviews were much more detailed and could point to more specific items since it wasn’t reviewing a summary of the work. I also did not have a problem where it said something like “you didn’t include x” when x was definitely stated out clearly in the work.” (Anonymous #6)

> “The AI review for this iteration impressed me with its increased depth and precision. It wasn't just highlighting areas for improvement anymore; it was offering constructive suggestions on how to address them. This actionable feedback proved invaluable in refining my work.” (Anonymous #22)

> “...the AI’s review of my paper has only furthered my confidence. It is crazy with how good this review was, and this was my second time using it for an [course name] course!” (Anonymous #29)

Quantitatively, the mean quality score for AI reviews was recorded at 3.13, slightly higher than that for human reviews, which averaged 2.80. Furthermore, AI-generated reviews were perceived as significantly more useful (Mean = 3.07) compared to human reviews (Mean = 2.38). Interestingly, the mean actionability scores for both AI-generated (2.67) and human-generated (2.66) reviews were similar. These findings significantly diverge from the results obtained in our second research question (RQ2). However, further rigorous testing and significance analysis are warranted, involving a larger sample size with sufficient statistical power to detect differences.

Building upon the nuanced insights gained from this study, the future of feedback mechanisms should navigate towards a harmonious integration of both peer and AI feedback. But, it might require a multifaceted approach to address existing research gaps, policy and ethical considerations, and implementation challenges. Longitudinal studies are warranted to gauge the sustained impact of integrated methodologies, moving beyond the current emphasis on short-term outcomes. Investigating the impact of blinding on participants' ability to differentiate between AI and human feedback could be crucial for understanding the dynamics of source transparency. Ethical considerations, encompassing privacy, bias mitigation, and responsible AI use in education, necessitate relevant policy changes and guidelines. Interdisciplinary collaboration, global perspectives, continuous iteration, and standardized evaluation frameworks can collectively contribute to shaping a comprehensive future for integrated feedback, fostering effective learning environments, and meeting the diverse needs of learners.

Limitations

While this research provides valuable insights into the impact of AI-assisted feedback, certain limitations warrant consideration. Firstly, the study heavily relies on self-reported data, potentially introducing response bias. Secondly, seasonal variations, academic workload, or other temporal factors (including agile research methodologies and pedagogical design) may differently influence participants' responses in other terms or years. Particularly, the selection of people into this sample can be an important caveat to the claims made, considering the diverse sample encompassing varied age groups, genders, educational backgrounds, and global locations. Even though this inclusivity contributes to a comprehensive exploration of perceptions, it may not be universally applicable to individuals with significantly different characteristics. Finally, the exclusive focus on short-term post-intervention outcomes raises questions about the sustained impact of the observed improvements over an extended duration, suggesting a need for future longitudinal investigations.

Conclusions

This study uncovers the pivotal role of emotions in shaping perceptions of AI-assisted feedback, reflecting the perspectives outlined by Reeves & Nass (1996). Positive emotions significantly enhance the perceived quality, usefulness, and actionability, while negative emotions, such as anxiety, reveal a substantial inverse relationship. These insights shed light on the intricate dynamics of user experiences in AI feedback scenarios. What’s more, peer ratings were consistently slightly higher than AI ratings across dimensions of quality, usefulness, and actionability, indicating a discernible preference for peer feedback aligning with the findings of Tzirides et al. (2023a) and Mayer et al. (2023). One potential reason for the consistently slightly higher peer ratings could be that peer feedback often involves a reciprocal exchange between individuals who share similar experiences and understand each other's challenges and perspectives. Additionally, peer feedback offers opportunities for dialogue and clarification that may not be present in AI-generated feedback, at least not during the iteration of this study. The key distinction arises from the fact that participants were not blind to the source of the feedback. This raises the possibility that the preference for peer feedback might be rooted in less likeness of receiving feedback from a computer, rather than an actual inferiority in the feedback provided. Even so, further exploration would be needed to uncover other explanations for this pattern of association.

Moreover, these findings highlight demographic nuances, emphasizing the need for tailored feedback mechanisms. In addition, the qualitative analysis also elucidated the operational disparities and strengths of peer and AI reviews perceived by the students, offering valuable insights into their respective advantages and shortcomings. Despite a clear preference for peer reviews, participants exhibit openness to AI feedback, recognizing its potential support and enrichment, aligning with the viewpoints presented by Zapata et al. (2024). This challenges the assumption of a general aversion to AI and suggests a nuanced perspective. Not to forget, the integration of advancement techniques (like RAG) into the LLMs represents a promising avenue for leveraging AI to enhance the learning experience and transform assessment. Based on our initial findings of the Spring 2024 implementation, by incorporating RAG techniques, AI-generated reviews have demonstrated their ability to provide clear, well-organized feedback that students find useful for their learning and improvement.

Consecutively, the assumption that blinding eliminates the ability to differentiate between AI and peer reviews, might hold some merit given the previous assumption was negated. However, further research would be warranted to confidently make that claim. Specifically, when participant's expressed preferences and reflections indicate an awareness and appreciation for the unique qualities of both AI and human evaluations, even when the source is concealed.

Yet, the diverse sample, encompassing varied age groups, genders, educational backgrounds, and global locations, significantly contributes to a comprehensive exploration of participants' perceptions of peer and AI feedback. Notably, it encompasses all graduate students enrolled in academic terms throughout 2023 (Spring, Summer, and Fall), offering a comprehensive and representative insight into the perspectives of students within a US-based university. This inclusive approach serves as a robust measure, capturing a holistic view of students' perceptions across semesters. The richness of this sample enhances the generalizability of findings across a broad spectrum of experiences and perspectives within the context of US-based higher education.

In conclusion, the juxtaposition of peer and AI reviews unveils the potential for a symbiotic integration of both approaches to optimize the feedback process. By harnessing the strengths of each method, we can cultivate a more comprehensive and effective feedback ecosystem, enhancing learning outcomes and fostering a culture of continuous improvement. Nonetheless, as we navigate the evolving landscape of educational technology, it is imperative to remain vigilant of the evolving landscape of educational technology and feedback mechanisms.

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Appendix

Appendix 1: IRB Document

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

Office of the Vice Chancellor for Research & Innovation

Office for the Protection of Research Subjects
1901 S. First St., Suite A, MC-685
Champaign, IL 61820

Notice of Exempt Determination

July 13, 2023

Principal Investigator	William Cope
CC	Akash Saini
Protocol Title	*Online learning experience: Using Scholar as an e-learning platform.*
Protocol Number	17887
Funding Source	Unfunded
Review Category	Exempt 2
Approval Date	July 13, 2023
Expiration Date	July 12, 2028

This letter authorizes the use of human subjects in the above protocol. The University of Illinois at Urbana-Champaign Office for the Protection of Research Subjects (OPRS) has reviewed your application and determined the criteria for exemption have been met.

The Principal Investigator of this study is responsible for:

- Conducting research in a manner consistent with the requirements of the University and federal regulations found at 45 CFR 46.
- Requesting approval from the IRB prior to implementing major modifications.
- Notifying OPRS of any problems involving human subjects, including unanticipated events, participant complaints, or protocol deviations.
- Notifying OPRS of the completion of the study.

Changes to an **exempt** protocol are only required if substantive modifications are requested and/or the changes requested may affect the exempt status.

Appendix 2: Post-Course Survey Questions

Questions	*Type*	*Response Options*
What is your age?	Single-select	• 18-24 years old • 25-35 years old • 35-45 years old • 45+ years old • Prefer not to say
What gender do you identify as?	Single-select	• Female • Male • Non-binary • Other • Prefer not to say
Where are you located?	Single-select	• Africa • Asia • Australia • Europe • North America • South America • Other
What degree are you pursuing?	Single-select	• Bachelor • Master's • Doctorate • Certificate • Non-degree • Other
How did you feel about having your work reviewed by an AI tool?	Single-select	• Uncomfortable • Anxious • Indifferent • Curious • Excited • Other
Which type of review would you prefer to receive?	Single-select	• AI review • Peer (Human) reviews • Both • Neither
How would you rate the overall quality of the AI review created?	Single-select	• Very Poor • Poor • Average • Good • Excellent
How useful or not did you find the AI review in improving your work based on the course rubric?	Single-select	• Not at all useful • Slightly useful • Moderately useful • Very useful

		• Excellently useful
How easy or difficult was understanding and implementing the AI reviews' suggestions in your work revision?	Single-select	• Very difficult • Somewhat difficult • Neither easy nor difficult • Somewhat easy • Very easy
List two advantages of the AI review you received.	Open-ended	
List two disadvantages of the AI review you received.	Open-ended	
How would you rate the overall quality of the peer (human) reviews recieved?	Single-select	• Very Poor • Poor • Average • Good • Excellent
How useful or not did you find the peer (human) reviews in improving your work based on the course rubric?	Single-select	• Not at all useful • Slightly useful • Moderately useful • Very useful • Excellently useful
How easy or difficult was understanding and implementing the peer (human) reviews' suggestions in your work revision?	Single-select	• Very difficult • Somewhat difficult • Neither easy nor difficult • Somewhat easy • Very easy
List two advantages of the AI review you received.	Open-ended	
List two disadvantages of the AI review you received.	Open-ended	

Appendix 3: Feeling Towards AI

Regression - Feeling of anxious and uncomfortable, and feeling of curious and excited

Source	SS	df	MS
Model	6.87891081	1	6.87891081
Residual	14.3278689	293	.048900576
Total	21.2067797	294	.072131904

Number of obs	=	295
F(1, 293)	=	140.67
Prob > F	=	0.0000
R-squared	=	0.3244
Adj R-squared	=	0.3221
Root MSE	=	.22113

dummy_anx_~m	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
dummy_ex_cur	-.3770492	.0317903	-11.86	0.000	-.4396155	-.3144828
_cons	.3770492	.0283134	13.32	0.000	.3213258	.4327726

ANOVA - Feeling of anxious and uncomfortable, and feeling of curious and excited

Number of obs =	295	R-squared =	0.3244
Root MSE =	.221135	Adj R-squared =	0.3221

Source	Partial SS	df	MS	F	Prob>F
Model	6.8789108	1	6.8789108	140.67	0.0000
dummy_ex_~r	6.8789108	1	6.8789108	140.67	0.0000
Residual	14.327869	293	.04890058		
Total	21.20678	294	.0721319		

Quality AI and Participants feeling curious or excited with AI review:

qualityAI	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
dummy_ex_cur	.5697072	.112054	5.08	0.000	.3491744	.7902399
_cons	2.131148	.0997986	21.35	0.000	1.934735	2.32756

Rating: 0= Very Poor, 1=Poor, 2=Average, 3=Good, 4=Excellent

Quality AI and Participants feeling anxious or uncomfortable with AI review:

qualityAI	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
dummy_anx_uncom	-.5851982	.1732261	-3.38	0.001	-.9261234	-.2442731
_cons	2.628676	.0483689	54.35	0.000	2.533482	2.723871

Rating: 0= Very Poor, 1=Poor, 2=Average, 3=Good, 4=Excellent

Usefulness AI and Participants feeling anxious or uncomfortable with AI review:

usefulAI	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
dummy_anx_uncom	-.8335997	.2103986	-3.96	0.000	-1.247684	-.4195157
_cons	2.268382	.0587483	38.61	0.000	2.15276	2.384005

Rating: 0= Not at all useful, 1=Slightly useful, 2=Moderately useful, 3=Very useful, 4=Excellently useful

Usefulness AI and Participants feeling curious or excited with AI review:

usefulAI	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
dummy_ex_cur	.6490822	.1378517	4.71	0.000	.3777773	.9203872
_cons	1.688525	.1227747	13.75	0.000	1.446892	1.930157

Rating: 0= Not at all useful, 1=Slightly useful, 2=Moderately useful, 3=Very useful, 4=Excellently useful

Actionability AI and Participants feeling anxious or uncomfortable with AI review:

actionAI	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
dummy_anx_uncom	-1.359335	.2092721	-6.50	0.000	-1.771202	-.9474681
_cons	2.794118	.0584338	47.82	0.000	2.679114	2.909121

Rating: 0= Very difficult, 1=Somewhat difficult, 2=Neither easy nor difficult, 3=Somewhat easy, 4=Very easy

Actionability AI and Participants feeling curious or excited with AI review:

actionAI	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
dummy_ex_cur	.8468544	.1396807	6.06	0.000	.5719498	1.121759
_cons	2.016393	.1244037	16.21	0.000	1.771555	2.261232

Rating: 0= Very difficult, 1=Somewhat difficult, 2=Neither easy nor difficult, 3=Somewhat easy, 4=Very easy

Appendix 4: Quality, Usefulness, and Actionability for AI and Peer Feedback

Paired t-test for Quality

Paired t test

Variable	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
qualit~I	295	2.583051	.0472604	.8117237	2.490039	2.676062
qualit~n	295	2.79661	.0540951	.9291145	2.690147	2.903073
diff	295	-.2135593	.0687174	1.18026	-.3487996	-.078319

```
     mean(diff) = mean(qualityAI - qualityhuman)                  t =  -3.1078
 H0: mean(diff) = 0                              Degrees of freedom =      294

 Ha: mean(diff) < 0           Ha: mean(diff) != 0           Ha: mean(diff) > 0
 Pr(T < t) = 0.0010         Pr(|T| > |t|) = 0.0021          Pr(T > t) = 0.9990
```

Paired t-test for Usefulness

Paired t test

Variable	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
usefulAI	295	2.20339	.0578045	.9928256	2.089627	2.317153
useful~n	295	2.60678	.0613453	1.05364	2.486048	2.727511
diff	295	-.4033898	.0811395	1.393616	-.5630776	-.2437021

```
     mean(diff) = mean(usefulAI - usefulhuman)                    t =  -4.9716
 H0: mean(diff) = 0                              Degrees of freedom =      294

 Ha: mean(diff) < 0           Ha: mean(diff) != 0           Ha: mean(diff) > 0
 Pr(T < t) = 0.0000         Pr(|T| > |t|) = 0.0000          Pr(T > t) = 1.0000
```

Paired t-test for Actionability

Paired t test

Variable	Obs	Mean	Std. err.	Std. dev.	[95% conf. interval]	
actionAI	295	2.688136	.0599116	1.029015	2.570226	2.806046
action~n	295	2.925424	.0515369	.8851752	2.823996	3.026852
diff	295	-.2372881	.0697039	1.197205	-.3744701	-.1001062

```
     mean(diff) = mean(actionAI - actionhuman)                    t =  -3.4042
 H0: mean(diff) = 0                              Degrees of freedom =      294

 Ha: mean(diff) < 0           Ha: mean(diff) != 0           Ha: mean(diff) > 0
 Pr(T < t) = 0.0004         Pr(|T| > |t|) = 0.0008          Pr(T > t) = 0.9996
```

Appendix 5: Influence of Age, Gender, and Eductaion on Perceived Quality, Usefulness and Actionability

Age and Quality:

Age Group		Obs	Mean	Std. dev.	Min	Max
18-24	AI	16	2.125	1.204159	0	4
	Human	16	2.6875	1.195478	0	4
25-35	AI	128	2.570313	0.7174275	1	4
	Human	128	2.78125	0.9301384	0	4
35-45	AI	83	2.614458	0.7782301	0	4
	Human	83	2.722892	0.9280619	0	4
45+	AI	65	2.707692	0.8229935	0	4
	Human	65	2.969231	0.8654701	0	4

Rating: 0= Very Poor, 1=Poor, 2=Average, 3=Good, 4=Excellent

Gender and Quality:

Gender		Obs	Mean	Std. dev.	Min	Max
Female	AI	202	2.643564	0.8114131	0	4
	Human	202	2.826733	0.9113318	0	4
Male	AI	84	2.5	0.7838736	0	4
	Human	84	2.797619	0.9285273	0	4

Rating: 0= Very Poor, 1=Poor, 2=Average, 3=Good, 4=Excellent

Education and Quality:

Education		Obs	Mean	Std. dev.	Min	Max
Bachelor	AI	0	-	-	-	-
	Human	0	-	-	-	-
Master's	AI	173	2.520231	0.8253935	0	4
	Human	173	2.791908	0.9166061	0	4
Doctorate	AI	115	2.678261	0.7897127	0	4
	Human	115	2.791304	0.9504907	0	4
Non-degree	AI	2	2	1.414214	1	3
	Human	2	3.5	0.7071068	3	4
Certificate	AI	3	2.666667	0.5773503	2	3
	Human	3	3.333333	1.154701	2	4

Rating: 0= Very Poor, 1=Poor, 2=Average, 3=Good, 4=Excellent

Age and Usefulness:

Age Group		Obs	Mean	Std. dev.	Min	Max
18-24	AI	16	2.1875	1.108678	0	4
	Human	16	2.75	1.437591	0	4
25-35	AI	128	2.125	0.9720504	0	4
	Human	128	2.554688	1.040961	0	4
35-45	AI	83	2.240964	0.9185137	0	4
	Human	83	2.542169	1.039479	0	4
45+	AI	65	2.353846	1.066987	0	4
	Human	65	2.784615	0.9436019	0	4

Usefulness Rating: 0= Not at all useful, 1=Slightly useful, 2=Moderately useful, 3=Very useful, 4=Excellently useful

Gender and Usefulness:

Gender		Obs	Mean	Std. dev.	Min	Max
Female	AI	202	2.282178	0.9745923	0	4
	Human	202	2.653465	1.055058	0	4
Male	AI	84	2.059524	1.045371	0	4
	Human	84	2.571429	1.009422	0	4

Usefulness Rating: 0= Not at all useful, 1=Slightly useful, 2=Moderately useful, 3=Very useful, 4=Excellently useful

Education and Usefulness:

Education		Obs	Mean	Std. dev.	Min	Max
Bachelor	AI	0	-	-	-	-
	Human	0	-	-	-	-
Master's	AI	173	2.115607	1.0164	0	4
	Human	173	2.589595	1.088687	0	4
Doctorate	AI	115	2.356522	0.9383758	0	4
	Human	115	2.634783	1.002894	0	4
Non-degree	AI	2	1	1.414214	0	2
	Human	2	3	1.414214	2	4
Certificate	AI	3	2	1	1	3
	Human	3	2.666667	1.527525	1	4

Usefulness Rating: 0= Not at all useful, 1=Slightly useful, 2=Moderately useful, 3=Very useful, 4=Excellently useful

Age and Actionability:

Age Group		Obs	Mean	Std. dev.	Min	Max
18-24	AI	16	2.375	1.087811	1	4
	Human	16	2.6875	1.138347	0	4
25-35	AI	128	2.828125	0.9809956	0	4
	Human	128	3.070313	0.8710894	0	4
35-45	AI	83	2.638554	1.019066	0	4
	Human	83	2.759036	0.9185137	0	4
45+	AI	65	2.569231	1.117604	0	4
	Human	65	2.923077	0.776766	2	4

Actionability Rating: 0= Very difficult, 1=Somewhat difficult, 2=Neither easy nor difficult, 3=Somewhat easy, 4=Very easy

Gender and Actionability:

Gender		Obs	Mean	Std. dev.	Min	Max
Female	AI	202	2.633663	1.043557	0	4
	Human	202	2.975248	0.8834471	0	4
Male	AI	84	2.797619	1.003365	0	4
	Human	84	2.797619	0.8887484	0	4

Actionability Rating: 0= Very difficult, 1=Somewhat difficult, 2=Neither easy nor difficult, 3=Somewhat easy, 4=Very easy

Education and Actionability:

Education		Obs	Mean	Std. dev.	Min	Max
Bachelor	AI	0	-	-	-	-
	Human	0	-	-	-	-
Master's	AI	173	2.589595	1.039514	0	4
	Human	173	2.913295	0.9203747	0	4
Doctorate	AI	115	2.86087	0.9901883	0	4
	Human	115	2.947826	0.8464941	0	4
Non-degree	AI	2	1.5	0.7071068	1	2
	Human	2	3	1.414214	2	4
Certificate	AI	3	2.333333	1.527525	1	4
	Human	3	2.666667	0.5773503	2	3

Actionability Rating: 0= Very difficult, 1=Somewhat difficult, 2=Neither easy nor difficult, 3=Somewhat easy, 4=Very easy